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Dynamic prediction and control of a tunnel boring machine with a particle swarm optimization–random forest algorithm and an integrated digital twin

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HIGHLIGHTS

- Research focuses on reducing the negative impact during shield tunneling construction process.
- The high-performance machine learning models is utilized to predict and adjust the attitude of TBM.
- The reasonable control range of shield construction parameters is obtained.
- A twin shield attitude control system is proposed based on digital twin technology.

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ABSTRACT

Tunnel boring machines (TBMs) often experience attitude deviation during excavation, impacting the stability and safety of tunnel construction. Traditional attitude adjustment relies on manual adjustment, which has a lagging effect. Therefore, the study combines a digital twin platform and a hybrid intelligence algorithm to enable real-time adjustment of the shield attitude deviation. The particle swarm optimization and random forest (PSO-RF) algorithm is first used to make accurate predictions of the shield attitude. Shapley additive explanations (SHAP) is subsequently employed to identify the key construction parameters. Then, based on these parameters, a control system for the shield attitude is designed in conjunction with a digital twin (DT) technique. A case study of China's Guiyang Metro Line 3 demonstrates the following: (1) The PSO-RF model achieves high accuracy, with R^2 values ranging from 0.916 to 0.943 for six shield attitude targets. (2) The key shield parameters are continuously optimized and adjusted within the control range to achieve shield attitude control. (3) The digital twin system provides real-time attitude warnings and parametric inference, significantly improving TBM performance and safety. In this paper, a novel method of combining predictive modeling and the DT platform is proposed. Under the proposed intelligent method, the attitude deviation of a TBM during tunneling was significantly reduced.

1. Introduction

With the growing demand for underground transportation systems, tunnel construction faces numerous challenges [1], particularly in maintaining shield attitude control. Shield tunnels have become the preferred method for underground tunnel construction because they adapt to diverse geological conditions. The shield construction process for rail transit is extremely complex and involves many construction techniques [2]. Shield attitude control while tunneling is key to ensuring sufficient tunnel construction quality [3,4]. During shield construction, the shield tunnel axis is affected by external factors such as excavation surface disturbances, shield quality changes and shield operation difficulties. As a result, the axis direction of the tunnel is difficult to control, and the tunnel can easily deviate from the designed route [5]; moreover, the shield advancing attitude can tilt up or become deviated or

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Nomenclature

Js-U	Jack stroke – Up, mm
Js-D	Jack stroke - Down, mm
Js-L	Jack stroke - Left, mm
Js-R	Jack stroke - Right, mm
Jt-U	Jack thrust – Up, kN
Jt-D	Jack thrust - Down, kN
Jt-L	Jack thrust - Left, kN
Jt-R	Jack thrust - Right, kN
Sti-U	Shield tail interspace - Up, mm
Sti-D	Shield tail interspace - Down, mm
Sti-L	Shield tail interspace - Left, mm
Sti-R	Shield tail interspace – Right, mm
Cep	Chamber earth pressure, bar
Rssc	Rotation speed of screw conveyor, (rpm)* 1000
Ct	Cutterhead torque, kN·m
Cs	Cutterhead speed, r·min-1
As	Advancing speed, mm·min-1
Gp	Grouting pressure, bar
Gt	Gross thrust, kN
Sm	Shield muck, m^3
Fa	Foam agent, m^3
Ifa	Internal friction angle, °
Sc	Soil cohesion, kPa
Tbd	Tunnel burial depth, m
Hdsi	Horizontal displacement of shield incision, mm
Vdsi	Vertical displacement of shield incision, mm
Hdst	Horizontal displacement of shield tail, mm
Vdst	Vertical displacement of shield tail, mm
Paa	Pitch attitude angle, $mm \cdot m^{-1}$
Haa	Horizontal attitude angle, $mm \cdot m^{-1}$

serpentine, which may cause ground collapse in severe cases, resulting in safety- and quality-related problems [6]. Therefore, close attention must be paid to TBM attitude control during excavation to mitigate these issues. Some typical engineering cases also illustrate the challenges for TBM attitude control. In March 2014, shield tunnel axis deviation occurred on the right line of a route from Huayang Station to Guangdu North Station, which is part of the Chengdu Metro Line 1 southern extension. In July 2016, deviation in the tunnel between Dajicun Station and Xinggui Road Station on Nanning Metro Line 3 occurred. Addressing the defects caused by these accidents incurred significant economic losses and postponed the construction period by two months. Given these challenges, developing intelligent and timely methods for shield attitude adjustment remains a crucial research focus.

Ensuring the quality of shield construction necessitates precise shield attitude adjustment and control. At present, this adjustment process is predominantly manual, but because this adjustment needs to occur at the onset of deviation, timely control is difficult [7]. Hence, leveraging actual engineering monitoring data to predict the shield attitude is a beneficial approach for efficient and timely adjustment and control. In the past few decades, many studies have explored attitude prediction in shield construction. These methods fall into two categories: empirical formulas and numerical simulations. In terms of empirical formulas, Festa et al. [8] constructed an empirical formula based on a mechanism data analysis of shield construction and actual shield driving processes to control attitude deviation during shield construction. Liu et al. [9] established acceleration, momentum, kinematic and orientation equations for a tunnel boring machine (TBM) and performed position prediction and adjustment. The above research was based on empirical formulas. Although such approaches are simple and fast, it is difficult to accurately control a TBM in complex terrain and on steeply curved sections during construction [10]. In terms of numerical simulation methods, Sun et al. [11] used finite element simulations to analyze the shield correction process, examining how correction moments influence shield attitude and assessing the effects of different soil properties and burial depths on both parameters. These studies were based on a traditional numerical simulation method, which has the advantages of reflecting the physical laws governing tunnel construction and simulating the experimental scenario with high accuracy. However, numerical simulation methods often overlook potential coupling relationships between factors influencing shield attitude, while also suffering from excessive assumptions, high computational costs, limited factor consideration, and an inability to perform dynamic real-time calculations—leading to relatively low accuracy in shield attitude analysis and prediction [12].

In recent years, mainstream prediction research has shifted toward machine learning (ML) algorithms, driven by their potent data analysis capabilities and capacity to thoroughly explore potential coupling relationships among various influencing factors [13–16]. Li et al. [17] developed a variety of intelligent algorithms to propose a prediction model for estimating the penetration rate of TBMs under different geological conditions and achieved good application results in actual projects. Harandizadeh et al. [18] proposed a hybrid intelligent model called ICA-ANFIS-PNN, which provides an accurate, effective and applicable technique for the performance prediction of TBMs. Parsajoo et al. [19] used an artificial bee colony algorithm to optimize the parameters of an adaptive neuro-fuzzy inference system (ANFIS) and accurately predicted the brittleness index of rock samples from a project in Malaysia. Bardhan et al. [20] used a novel ensemble strategy based on input–output amalgamation techniques to predict the performance of TBMs and verified its effectiveness in a project conducted in Turkey. Xu et al. [21] constructed a database based on a project in Malaysia and accurately predicted the penetration rate of TBMs through supervised ML techniques. These studies demonstrate the potential of ML models for application in real engineering projects.

Recent years have seen extensive research on shield attitude prediction using ML methods, yielding significant results due to their

Table 1
Research on data-driven shield attitude prediction methods.

Literature	Algorithms	Prediction result
Hu et al. [22]	XGBoost	The R^2 and MSE for predicting the shield tunnel attitude are 0.753 and 9.03 mm, respectively, which can meet actual project requirements.
Chen et al. [23]	Bayesian optimization: LGBM	The R^2 values predicted for the six shield attitude parameters all reach 0.9 or above.
Zhang et al. [24]	Principal component analysis (PCA) and a gated recurrent unit (GRU)	The proposed framework enables real-time prediction of EPB shield tunneling trajectories.
Fu et al. [25]	PCA and time convolutional network (TCN)	The average prediction accuracy for four shield attitude parameters is 89.68 %.
Wang et al. [26]	Convolutional GRU neural network	In the advanced prediction of shield attitude, the highest R^2 is 0.9652, and the average R^2 is 0.9004.
Chen et al. [27]	Time-aware long-short-term memory (LSTM)	The model achieves R^2 values of 0.84 (HDSH), 0.97 (VDSH), 0.91 (HDST), 0.98 (VDST), 0.93 (ROLL), and 0.98 (PITCH).
Chang et al. [28]	Attention-based LSTM model	In the selected dataset, the highest R^2 for predicting the shield attitude reached 0.8526.

inherent advantages. Table 1 summarizes the recent progress in shield attitude prediction. ML methods can overcome the inherent drawbacks of empirical formulas and numerical simulations, thereby enabling precise predictions of the shield attitude. However, previous studies have focused mainly on predicting the shield attitude, with few reports on further regulating shield alignment on the basis of these predictions. In practical engineering cases, effectively regulating shield alignment on the basis of predictions is another issue that deserves close attention.

To adjust attitude in shield tunneling, many scholars have adopted different methods of identifying key operating parameters of shield tunneling and have adjusted the shield attitude on the basis of the identified operating parameters. After shield attitude prediction is performed, the effective adjustment and control of the shield attitude are vital for guiding engineering practice. Many researchers have reported that adjusting construction parameters is an effective means of controlling the shield attitude and ensuring construction safety [29]. Xie et al. [30] adjusted the shield attitude by changing the thrust distribution in the jack working area. Wang et al. [31] introduced a multicylinder shield control system that operates according to a leader–follower control strategy, where the system manages the shield attitude by manipulating the length of each motion chain. Wu et al. [32] established a connection between the shield jack oil pressure and shield attitude via historical engineering data through an ML algorithm, allowing for the control of the shield attitude by adjusting the oil pressure. Pan et al. [33] proposed an AutoDL framework for the dynamic prediction and multiobjective optimization of the TBM excavation attitude, which increases the reliability and efficiency of tunnel excavation. The adjustment of the shield tunneling attitude depends on precise predictions. However, practical engineering cases often pose challenges for shield attitude control because of the uneven mass distribution, operational complexities, and varying frictional resistance across sections. In addition, current shield attitude control methods rely on the experience of the shield driver, introducing a lag effect in the manual adjustment process. The shield needs to operate for a certain distance or time before the machine's motion path changes. To enable intelligent real-time shield attitude adjustment, an algorithm-physical interaction platform must be developed to enhance TBM excavation attitude monitoring and control.

In summary, the main research issues are as follows: (1) How can an index system for the influencing factors of the shield attitude be established, and on this basis, how can data be collected and processed to obtain valid data sample sets? (2) How can the shield attitude and nonlinear mapping relationship between shield parameters be established to quickly and accurately predict the shield attitude? (3) How can the importance of shield parameters be evaluated, the key control parameters in construction be defined, and effective shield attitude control be realized? This research incorporates a digital twin model for the real-time monitoring of the TBM excavation posture and utilizes a particle swarm optimization and random forest (PSO-RF) model to map the complex relationship between the operational parameters and the TBM posture. In addition, the parameter control range for the effective adjustment of the TBM excavation posture under local conditions is identified. This paper seeks to provide a new paradigm for collaborative decision making between operators and intelligent algorithms to provide a feasible method for adaptive shield attitude control.

The article is structured as follows. Section 2 summarizes the research significance of this paper. Section 3 describes the proposed shield attitude prediction and control framework. Section 5 uses a case study to show the effectiveness of the proposed method. Section 6 compares the proposed method with the latest approaches. Section 7 presents the limitations and future research directions, and Section 7 concludes the paper.

2. Research significance

Existing methods have yielded significant results in predicting changes in the attitudes of shield machines, thereby providing guidance for the on-site construction of shield tunnels. However, prior research has focused predominantly on predicting shield attitudes, while neglecting the subsequent control based on these predictions. When a shield machine initiates excavation, changes in its attitude arise from complex interactions between the machine and the surrounding ground, which are influenced by multiple parameters, including the shield parameters and geological properties. This dynamic process is highly intricate, rendering precise analysis through theoretical and numerical methods very challenging. Although numerous ML methods have been employed to predict issues related to attitude changes during shield excavation, limited research has identified key parameters based on these predictions, and no study has yet established a collaborative decision-making mechanism that integrates intelligent algorithms with manual operations.

A new approach is necessary to more effectively predict and provide just-in-time control of the changes in the shield attitude during excavation. Therefore, this paper proposes a dynamic prediction and control method for shield attitudes, which integrates a high-performance PSO-RF model and a digital twin platform. The key contributions include:

(1) **Effective fusion of data from multiple sources.** By analyzing the mechanisms of shield attitude deviation, this study identifies influencing factors, constructs a comprehensive index system, and preprocesses the data to create an effective sample dataset.

(2) **Establishment of high-performance predictive models.** An ML algorithm is employed to achieve accurate shield attitude prediction, with the PSO-RF algorithm validated for its effectiveness in model construction.

(3) **Exploration of collaborative decision-making mechanisms between operators and intelligent models.** Using the SHAP method, key parameters significantly impacting shield attitude are identified. These parameters are then dynamically adjusted and optimized within a digital twin framework to guide real-time attitude control.

In summary, this study integrates digital twin technology with shield attitude adjustment processes to design a control system. The proposed framework not only improves predictive accuracy but also identifies critical control parameters, offering a solution for intelligent shield attitude management.

3. Methodology

Due to the complexity of TBM operations and their working environment, multiple coupling factors make it challenging to establish precise theoretical descriptions or numerical models. To address this, this study proposes a shield attitude prediction model using a PSO-RF algorithm, with attitude control achieved through iterative adjustments of key construction parameters. The implementation of this approach is based on the Python platform; the detailed flowchart is given in Fig. 1. The approach has three main steps: data acquisition and preprocessing, shield attitude prediction, and key construction parameter optimization. The specifics of each step are discussed in detail in the following sections: 3.1, Data collection and preprocessing; 3.2, Development of the shield attitude prediction model; and 3.3, Shield attitude control on the basis of key construction parameters.

3.1. Data collection and preprocessing

The sensors embedded in the TBM provide real-time monitoring of operational status during excavation. However, some of the collected data values will be abnormal, and this will have a negative impact on the modeling of shield posture prediction. Consequently, data preprocessing is conducted to extract valuable information and transform it into a format suitable for model development. This preprocessing involves outlier cleaning, filling, normalization, and Pearson correlation analysis.

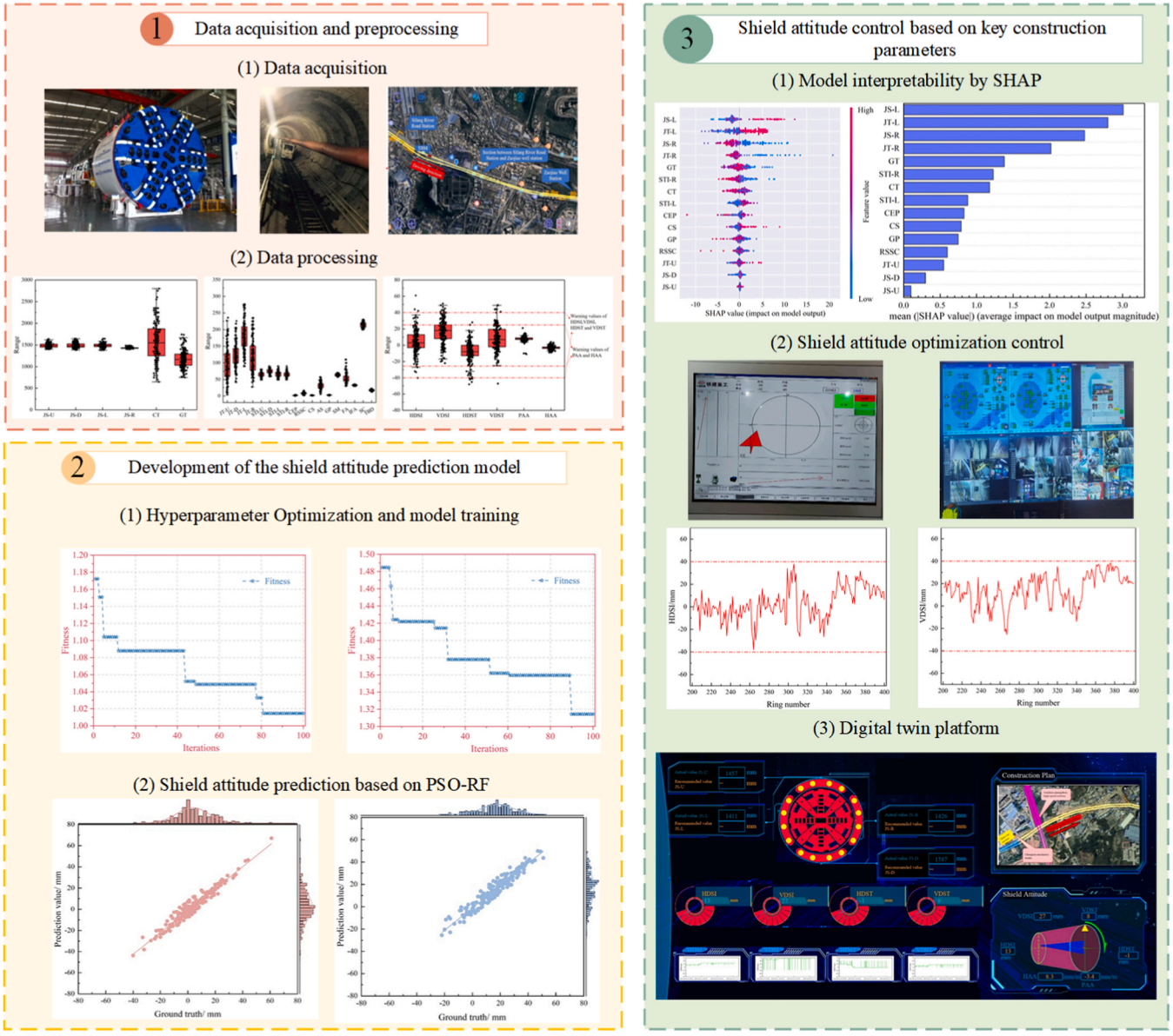


Fig. 1. Framework of the proposed shield attitude control method.

3.1.1. *Outlier cleaning*

Because the sensors and data acquisition instruments used in tunneling systems are always operating, the data collected include not only the data acquired under normal shield working conditions but also those acquired during shutdowns. The data from the shutdown state have no reference value regarding the operating state in the tunneling system, and the data from the shutdown and normal operation states are generally mixed together, which greatly affects model construction and accuracy. To address this, an abnormal data filter is introduced to improve analysis precision. Abnormal data can be categorized into nonworking shutdown data and abnormal data. These two types of data need to be identified and treated in different ways.

The data acquired in a shutdown state are collected mainly when the cutterhead is replaced or when the shield is stopped or maintained. The nonworking state of a TBM can be identified according to the relevant construction parameters. In this work, four major construction parameters are selected as indices: the gross thrust, cutterhead torque, advance speed and cutterhead speed. When one of the four construction parameters is equal to zero, the shield can be considered to be in a nonworking state. The corresponding discriminant function is as follows [34]:

$$D = F{\cdot}T{\cdot}v{\cdot}n \tag{1}$$

$$Status = \begin{cases} Normal\ working\ condition\ ,\ D \neq 0 \\ Abnormal\ working\ condition\ ,\ D = 0 \end{cases} \tag{2}$$

where F, T, v and n correspond to the gross thrust, cutterhead torque, advance speed and cutterhead speed, respectively, and D represents the product of these four parameter values.

During data acquisition, electromagnetic noise and strong mechanical vibrations at the construction site introduce significant anomalies, causing large deviations in subsequent analysis. To ensure reliable model training and prediction, these abnormal data must be filtered out. The penetration degree is often used as an indicator of the working state of a cutterhead. This paper adopts the penetration degree as the criterion for identifying abnormal data in a normal working state. The formula for penetration P is shown in Eq. (3). The rings refer to the number of each ring excavated by the shield machine in underground tunnel engineering and are typically counted from the starting point of the tunnel. Fig. 2 shows the penetration data for 201–400 rings.

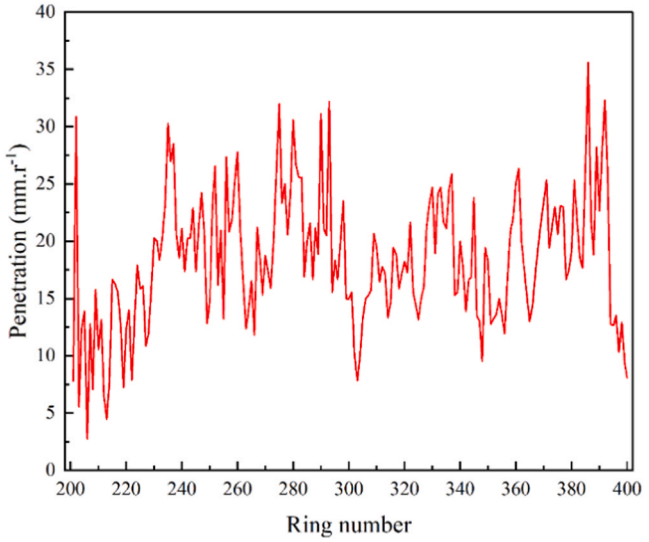


Fig. 2. Penetration degrees of 201–400 rings.

$$P = v/n \tag{3}$$

As shown in Fig. 2, the first three rings of penetration have an obvious peak, and the rapid increase in penetration is dramatically reduced again because the shield construction machine operates in a state of flux in the initial stages. The data collected at this time are not meaningful and will disrupt subsequent data analysis. Consequently, it is necessary to distinguish between data from the initial working stage and those from the normal working state and eliminate the data in the initial working stage.

In the normal working state, there are also several abnormal data points, which are characterized by abnormally large or small penetrations. Therefore, the penetration degrees observed for the 200 rings are grouped for statistical analysis, and abnormally large or small penetration degrees are excluded.

3.1.2. Data filling

When outliers and data in the initial working stage are removed from the collected data, the resultant gaps need to be filled. Common filling methods include statistical filling, filling on the basis of correlation attributes and interpolation filling.

In this paper, the cubic interpolation method is used to fill these gaps. For the function $S(x) \in C^2[a,b]$, considering a cubic polynomial in the interval $[x_{i-1}, x_i](i = 1, 2, \ldots, n)$ and assuming $a = x_0 < x_1 < \ldots < x_{n-1} < x_n = b$, if a function $y_i = f(x_i)(i = 1, 2, \ldots, n-1)$ is given for node x_i and the condition $S(x_i) = y_i(i = 0, 1, \ldots, n)$ is satisfied, then $S(x)$ is a cubic spline interpolation function, and its mathematical expression is [35]:

$$S(x) = M_{i-1}\frac{(x_{i-1})^3}{6h_{i-1}} + M_i\frac{(x - x_{i-1})^3}{6h_{i-1}} + \left(y_{i-1} - \frac{M_{i-1}h_{i-1}^2}{6h_{i-1}}\right)\frac{x_{i-1} - x}{h_{i-1}} + \left(y_i - \frac{M_i h_i^2}{6}\right)\frac{x - x_{i-1}}{h_{i-1}} \tag{4}$$

where $[x_{i-1}, x_i]$ is an intercell of $[a, b]$, x is a value in the interval $[x_{i-1}, x_i]$, h_{i-1} is the difference between x_i and x_{i-1}, and M_i is the second derivative of $S(x)$.

Notably, the geological parameter information obtained from core drilling sampling is usually discontinuous. In contrast, the operating parameter data collected by the shield machine sensor are continuous, and each segment has corresponding values. Therefore, this study uses the ordinary kriging method to expand the geological information so that each segment of the operation dataset has corresponding geological data.

3.1.3. Data normalization

Before training the model, normalizing data within the sample set is advantageous as it minimizes the effect of dimensional fluctuations on prediction efficiency and precision. Some studies indicate that scaling data samples to intervals of [-1, 1] can allow input variables to be directly compared and can reduce the potential impacts of different target amplitudes on subsequent optimization [36].

$$x_n = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \times (b - a) + a \tag{5}$$

where x_n is the normalized standard value. $x_{\max}$ is the maximum, and $x_{\min}$ is the minimum. a and b represent the normalized minimum and maximum values of the input variable x, respectively, and the initial values for a and b are -1 and 1, respectively.

3.1.4. Pearson correlation analysis

To ensure optimal model performance, it's crucial to perform feature selection and analyze inter-feature correlations beforehand. The Pearson correlation coefficient, calculated using Eq. (6) [37], serves as a standard measure for assessing linear relationships between variables.

$$R = \frac{N\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{N\sum x_i^2 - (\sum x_i)^2}\sqrt{N\sum y_i^2 - (\sum y_i)^2}} \tag{6}$$

where R is the Pearson correlation coefficient; x_i and y_i are the ith values of X and Y, respectively; and N is the number of samples.

3.2. Development of the shield attitude prediction model

To achieve accurate just-in-time prediction for shield attitude changes during excavation, a PSO-RF model was developed. The flowchart of the PSO-RF model for prediction and evaluation is illustrated in Fig. 3.

3.2.1. Random forest

RF is a powerful ensemble learning method that integrates statistical principles with ML, offering high predictive accuracy, robustness against overfitting, and strong generalization capabilities [38]. A repeated-sampling approach is adopted for RF based on a self-help method. The predictor $h(x, \theta_k)$ is subsequently constructed, and K classification trees are generated according to the self-help sample set to form an RF . The average predicted value of all the trees is used as the

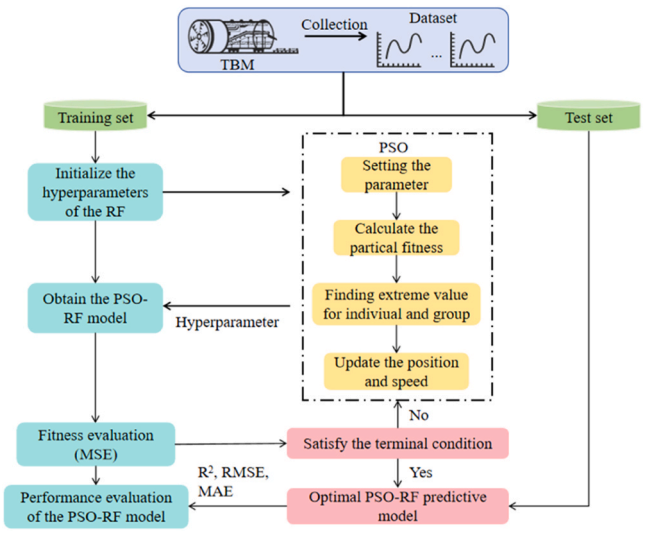


Fig. 3. Flowchart of the PSO-RF model.

final prediction result. The prediction result $h_i(x)$ of predictor $h(x,\theta_k)$ for each decision tree is recorded to obtain the prediction set $\{h_1(x), h_2(x), h_3(x), \ldots, h_k(x)\}$, and the final prediction result $h(x)$ can be expressed as [39]:

$$h(x) = \frac{1}{k}\sum_{i=1}^{k} h_i(x) \tag{7}$$

3.2.2. Particle swarm optimization algorithm

The PSO algorithm is utilized to determine the optimal hyperparameters for the ML algorithm. While alternative parameter optimization methods such as gray wolf optimization (GWO) and the whale optimization algorithm (WOA) can also be employed to find the optimal hyperparameters for the RF algorithm, these methods have comparable prediction accuracies [40]. In addition, the PSO algorithm stands out because of its simplicity, ease of implementation, minimal need for parameter adjustments, rapid convergence, low memory demands, and ability to avoid local optima.

In a D-dimensional space with a population size of n particles, each particle's current position represents a candidate solution in the optimization. Particle searches are iteratively performed until the optimal solution satisfying the termination condition is obtained. During each iteration, every particle updates its position based on its historical optimal position (P_{best}) and the optimal position in the population (G_{best}). The updating formula is expressed as follows [41]:

$$V_{id}^{k+1} = \omega V_{id}^{k} + C_1 r_1 \left(P_{id}^{k} - X_{id}^{k}\right) + C_2 r_2 \left(P_{id}^{k} - X_{id}^{k}\right) \tag{8}$$

$$X_{id}^{k+1} = X_{id}^{k} + V_{id}^{k+1} \tag{9}$$

where w denotes the inertial weight; d represents the dimension of the particle and is in the range $[1, D]$; i denotes the current ith particle and is in the range $[1, n]$; k signifies the current iteration number; V_{id} represents the velocity of the particle; X_{id} corresponds to the position of the particle; and C_1 and C_2 are learning factors. r_1 and r_2 are random numbers in the range [0,1].

3.2.3. Model performance evaluation

This study evaluates model performance using three key metrics: root-mean-square error (RMSE), mean absolute error (MAE) and goodness of fit (R^2). These indicators quantify prediction errors and model fit quality. Lower RMSE and MAE values reflect better accuracy, while an R^2 approaching 1 indicates stronger predictive capability. The equations for each index are presented below.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\mathrm{Pre}_i - Obs_i\right)^2} \tag{10}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|\mathrm{Pre}_i - Obs_i\right| \tag{11}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(Obs_i - \mathrm{Pre}_i\right)^2}{\sum_{i=1}^{n}\left(Obs_i - Obs_{ave}\right)^2} \tag{12}$$

where n denotes the sample number, Pre_i is the predicted value, Obs_i is the actual value, and Obs_{ave} represents the average of the actual values.

3.3. Shield attitude control based on key construction parameters

To enhance transparency in shield attitude control, SHAP additive interpretation was applied to the PSO-RF model, mitigating blind control risks and offering parameter insights for a digital twin platform. On this basis, this work further establishes a digital twin platform to strengthen the integrated control of TBM attitude prediction and regulation.

3.3.1. SHapley additive explanations

Owing to the "black box effect", the prediction results of the ML model often lack sufficient credibility [42]. To address this issue, many methods (such as tree-based methods, PCA and SHAP) have been used to interpret ML models. Among them, SHAP analysis has unique advantages. Compared with other methods, SHAP provides a consistent interpretation, and even if the predictions of the model change, the interpretation of the SHAP values should be remain stable as long as the characteristics of the feature contributions remain the same. In the study of shield attitude, these advantages of SHAP can make it easier to identify key shield parameters after precise predictions are made by the developed PSO-RF model. The SHAP method leverages game theoretic principles to assess the impacts of features on the model's output by calculating the Shapley value's marginal contribution for each feature. These contributions can be positive, indicating improved prediction results, or negative, signifying a reduction in prediction performance [43]. The formula for calculating $f(x_{ij})$ is as follows:

$$f(x_{ij}) = \sum\nolimits_{S\subseteq F\setminus\{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!}\left[v(S\cup\{j\}) - v(S)\right] \tag{13}$$

where $f(x_{ij})$ is the SHAP value of the j-th feature x_{ij} of the i-th sample x_i; F represents all feature sets in sample x_i; S represents the feature subset; $v(S)$ represents the contribution of the influence factors included in subset S; and $v(S\cup\{j\}) - v(S)$ represents the contribution of influencing factor j to the interaction. The cumulative mean of the influencing factors j for all samples is the SHAP value of the influencing factor j. The formula is as follows:

$$f(x_j) = \sum\nolimits_{i=1}^{M} f(x_{ij}) \tag{14}$$

3.3.2. A digital twin system for driving shield attitude control

The sensors installed on the TBM monitor and record real-time excavation data, which is processed by the onboard acquisition system. These records—containing shield parameters and excavation status—form the foundation for a three-dimensional digital twin platform. Physical equipment can be converted into digital models, which should reflect the actual characteristics of the modeled TBM. By combining the digital twin model with real-time monitoring data from the TBM and integrating intelligent modules, reliable estimates of the TBM excavation posture can be obtained.

By leveraging engineering monitoring data, the shield attitude can be accurately predicted through a mapping relationship between the target attitude and influencing factors. This prediction capability enables shield attitude control. Through SHAP analysis, considering the project's actual conditions and control requirements, key construction parameters during shield tunneling are selected for optimization and adjustment. This iterative process continues until the shield attitude conforms to the defined control specifications, ensuring the safety of tunneling operations. The twin shield attitude control system is subsequently created from the shield attitude control method and digital twin technology. Serving as a valuable addition to the existing shield operating system, the digital twin model significantly increases the intelligence and automation of shield attitude control. This approach enables shield attitude control on the basis of dual physical and data inputs, providing a comprehensive solution.

4. Case Study

In this study, real data from the Guiyang rail transit line are collected to validate the proposed method. First, based on mechanistic analysis and engineering experience, the parameters significantly influencing the shield attitude are identified. Then, a PSO-RF hybrid model is established to accurately predict changes in the shield attitude. Finally, the shield attitude is dynamically controlled by integrating the PSO-RF model with a digital twin (DT) platform.

4.1. Background

Guiyang Metro Line 3 construction commenced in late 2018, with an estimated total investment of 32.231 billion yuan. This line is an important transportation route for promoting the balanced and coordinated development of the northern and southern regions of the city and for achieving green, low-carbon transportation and environmental protection in the western urban agglomeration. On this line, the tunnel from the Sifanghe Road Station to Zaojiaojing Station (SZ) spans YDK24 + 511.010 to YDK25 + 628.283, covering a total distance of 1117.209 m. It features a maximum longitudinal slope of 25.144 ‰ and burial depths ranging from 12.69 to 21.72 m. Shield construction in this region employs an earth pressure balancing (EPB) TBM with a diameter of 6470 mm. The EPB-TBM is equipped with numerous sensors that continuously monitor its operating status. All operating parameter data are collected and recorded by the data acquisition system. The tunnel in this area mainly passes through medium-weathered dolomite that is partially insulated and partially hydrated, leading to complex hydrogeology. The dolomite in this region has a compressive strength of 45–78 MPa, a soil cohesion of approximately 200 kPa, and an internal friction of approximately 30°. Shield tunneling in such locally hydrated and complex formations complicates shield control. The shield tunneling process in rail transit construction is highly complex and requires precise control of the shield attitude within the challenging environment of the SZ construction zone. The project plan map, the composite EPB-TBM used in the project, the construction situation inside the tunnel and the interface of the shield attitude monitoring system are shown in Fig. 4(a)~(d), respectively.

4.2. Selecting the input parameters

The sensors equipped in the EPB-TBM can monitor real-time excavation conditions, with collected data processed by the onboard system and exported as CSV files organized by ring number. These files contain critical parameters including the excavation date and time, ring number, construction parameters, navigation parameters, etc. Based on literature review, key input parameters were selected to optimize model training efficiency, avoiding redundancy. The bottom of the cylinder of the shield thrust system rests on the lining segment of the upper ring, the jacking force generated by the segment is used for the segment, and the

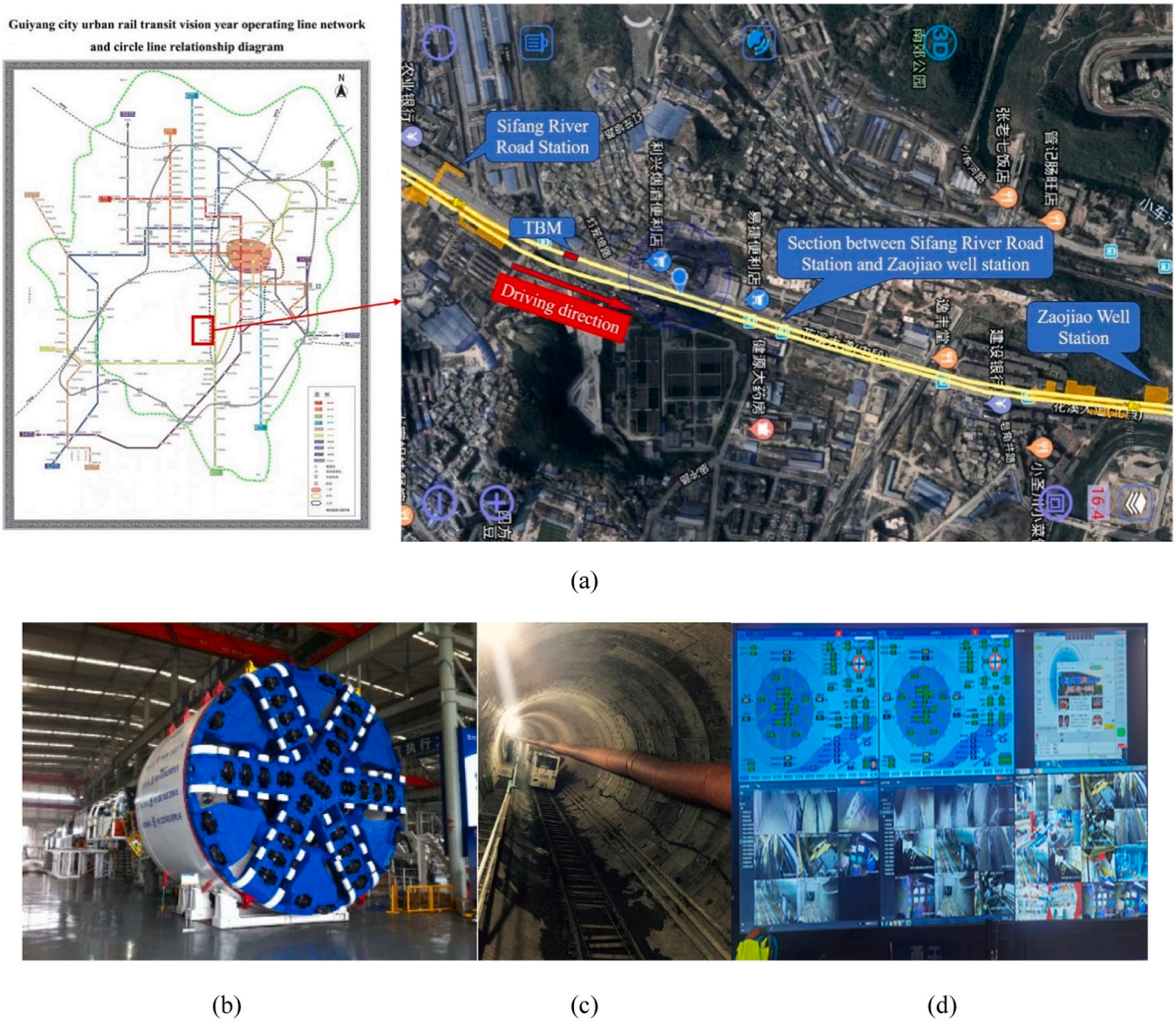


Fig. 4. Project overview: (a) Plane location map of the 7 standard projects of Guiyang Rail Transit Line 3; (b) composite EPB-TBM for the project; (c) construction of the interior of the tunnel; (d) interface of the shield attitude monitoring system.

reaction force generated by the segment drives the TBM forward. The jack stroke and thrust are two key parameters that directly affect the shield attitude [44]. One of these key parameters is the shield tail clearance. Inadequate clearance may cause the shield to apply excessive pressure on the segment, risking dislocation or cracking [45]. Maintaining equilibrium between chamber earth pressure and tunnel face pressure is crucial for minimizing soil disturbance, making chamber earth pressure a vital element in shield attitude control [46]. Controlling the rotation speed of the screw conveyor regulates soil discharge, maintaining this pressure balance [47]. Appropriate cutterhead torque and speed enhance cutting efficiency and mitigate instability during construction [48]. An increase in the advance speed contributes to greater soil disturbance, which affects the shield attitude [23]. Tail void grouting plays a crucial role in shield construction because it directly influences the load on the surrounding soil, thereby impacting the shield attitude during construction [49]. The thrust system, a vital shield subsystem, controls the shield attitude and position, with gross thrust being crucial for attitude control during construction [50]. For tunnel face stability, balancing soil discharge and excavation to maintain a stable support pressure is essential [51]. A foaming agent can regulate soil fluidity to reduce cutterhead torque and protect the shield cutterheads [52].

Moreover, numerous factors influence the shield attitude, including geological parameters and tunnel burial depth. Geological parameters reflect soil pressure in the shield environment. During shield attitude adjustment, the TBM interacts with surrounding soil mainly via shear and compression forces [53]. As the tunnel burial depth increases, the thrust of the TBM needs to increase to overcome the soil pressure, which poses certain challenges for the stability of the shield attitude [54].

In this paper, an input index system for shield attitude is introduced based on the previous mechanism analysis. The identified indices include jack stroke (JS), jack thrust (JT), shield tail clearance (STC), chamber earth pressure (CEP), rotation speed of the screw conveyor (RSSC), cutterhead torque (CT), cutterhead speed (CS), advance speed (AS), grouting pressure (GP), gross thrust (GT), shield muck (SM), foam agent (FA), internal friction angle (IFA), soil cohesion (SC), and tunnel burial depth (TBD). Among these, JS, JT, CEP, RSSC, CT, CS, AS, GP, and GT are the TBM's active control parameters that can be directly adjusted by operators. Conversely, STC, SM, and FA are passive control parameters beyond the operators' direct control. HDSI, VDSI, HDST, VDST, HAA, and PAA serve as descriptors of the shield's position state and are the most critical control indicators for shield operation, as shown in Fig. 5. Consequently, these variables are selected as the output variables for the prediction model. A comprehensive dataset for rings 201–400 was generated during TBM operation and collected, comprising data for a total of 200 rings.

4.3. Data preprocessing

Owing to shield shutdown, sensor signal interference and other reasons, many abnormal and invalid values exist in the original dataset, and they need to be preprocessed for subsequent analysis. In accordance with the process described in Section 3.1, data cleaning, data filling, data normalization and Pearson correlation analysis are successively carried out for the original data.

First, according to Eq. (1) and (2), we identify the shutdown data in the original dataset and eliminate these data directly. The penetration values are calculated according to the penetration Eq. (3), and the observed penetration ranges from 3 mm·r^{-1} to 36 mm·r^{-1}. The penetration degree was determined according to the obtained numerical values, and the distribution of the penetration degree was plotted. The statistical results are shown in Fig. 6(a), with T representing the data count and A representing the cumulative data percentage. The figure clearly shows that the distribution of the penetration degree closely resembles a normal distribution. Notably, data within the range of 3–30 mm·r^{-1} make up a substantial 96 % of the dataset. On the basis of these findings, combined with engineering experience, the data corresponding to $P > 30$ are eliminated. Moreover, the dataset is divided into an initial tunneling section under normal tunneling conditions and a stable tunneling section. The data pertaining to the initial tunneling section are subsequently excluded. After the resulting data gaps are filled via cubic spline interpolation, the penetration degrees of rings 201–400 are determined, as shown in Fig. 6(b). This figure demonstrates effective control of the penetration peak, yielding data representative of the normal working stage. In this paper, 200 data are used for model training and prediction, which reflect the operating behavior of shield drivers and the shield attitude changes. The statistical information for the input and output parameters after pretreatment is presented in

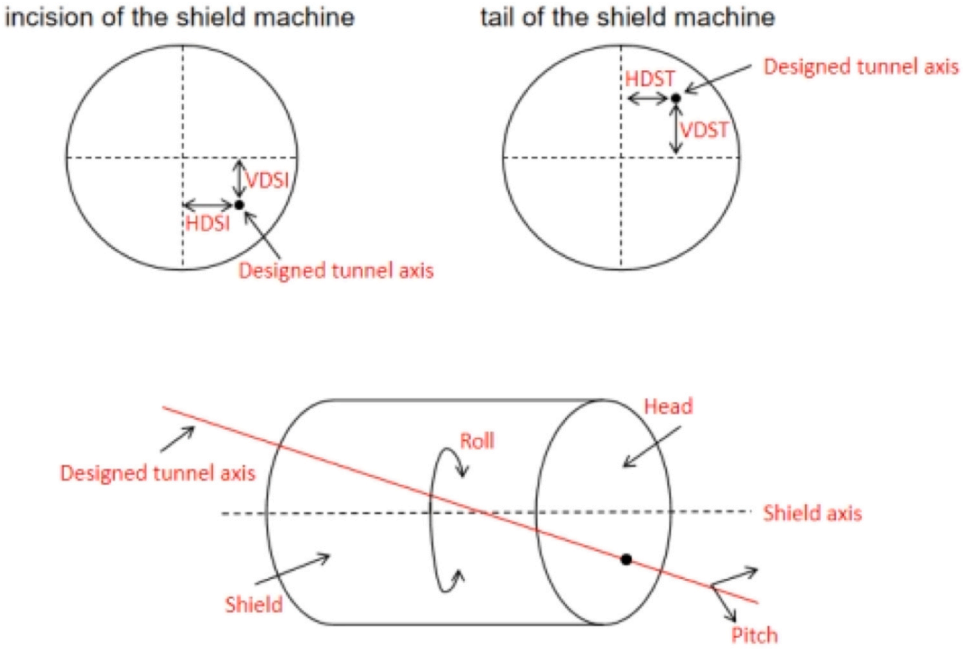


Fig. 5. Attitude parameters of the TBM.

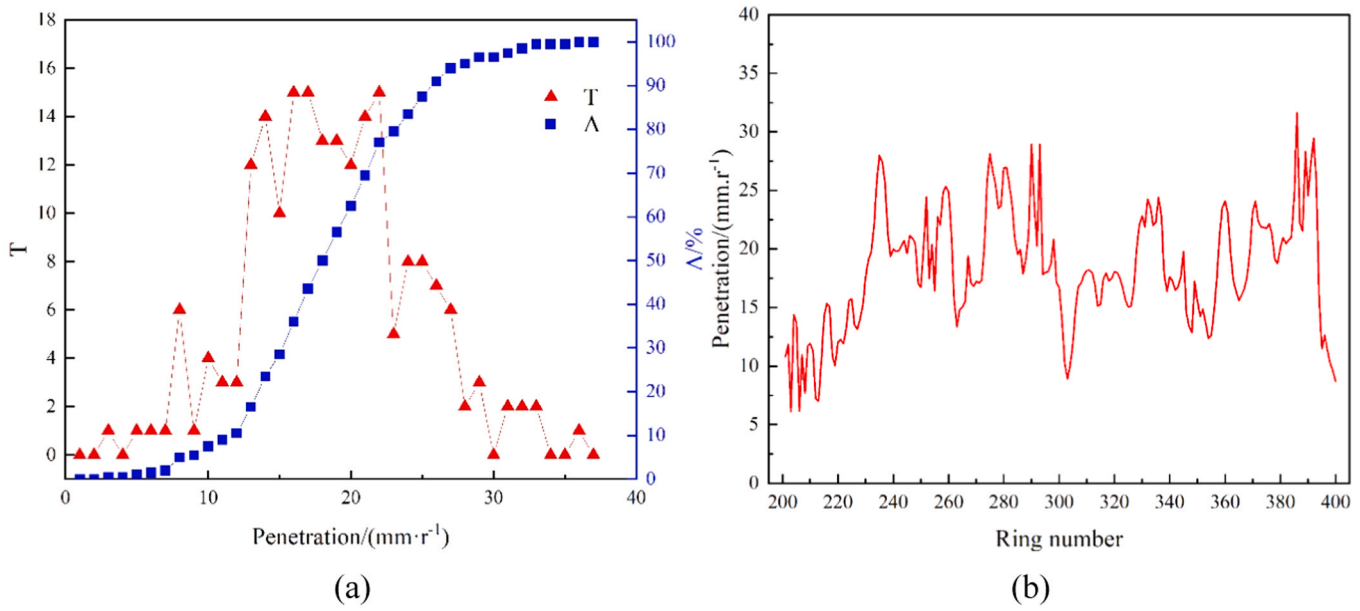


Fig. 6. Penetration: (a) the penetration distribution of the collected data; (b) the penetration degrees of rings 201–400 after data preprocessing.

Table 2
Input and output parameter data information.

Parameter	Variables	Parameter type	Data			Units
			Min	Ave	Max	
Shield parameter	JS-U	Input parameter	1634	1488	1407	mm
	JS-D		1644	1492	1387	mm
	JS-L		1645	1493	1383	mm
	JS-R		1485	1433	1404	mm
	JT-U		300	98	4	kN
	JT-D		226	124	42	kN
	JT-L		276	180	25	kN
	JT-R		235	117	27	kN
	STI-U		80	64	6	mm
	STI-D		580	79	60	mm
	STI-L		90	67	7	mm
	STI-R		550	68	50	mm
	CEP		3.3	1	0.2	bar
	RSSC		17	8	2	(rpm)* 1000
	CT		2800	1573	650	kN·m
	CS		2.9	2	1.3	$r \cdot min^{-1}$
	AS		58	32	5	$mm \cdot min^{-1}$
	GP		3.2	3	0.7	bar
	GT		9319	1209	126	kN
	SM		70	63	58	m^3
Soil property parameter	FA		150	56	30	m^3
	IFA		35	33	30	°
	SC		230	214	200	kPa
Tunnel design parameter	TBD		21.72	16.97	12.70	m
Shield attitude parameter	HDSI	Output parameter	61	4	−40	mm
	VDSI		51	17	−22	mm
	HDST		28	−8	−48	mm
	VDST		49	8	−47	mm
	PAA		21	8	−11	$mm \cdot m^{-1}$
	HAA		4	−3	−9	$mm \cdot m^{-1}$

Table 2, and a box diagram of the input and output variables is shown in Fig. 7.

Pearson's correlation coefficient analysis effectively assesses input - output parameter correlations. Fig. 8 presents the input - output parameter correlation heatmap. Input parameter correlation coefficients mostly fall within [-0.5, 0.5], suggesting weak, independent relations. The correlation coefficients between JS-U, JS-D, JT-U, JT-D, STI-U, and STI-D and PAA, VDSI and VDST are no greater than 0.8. The correlation coefficients between JS-L, JS-R, JT-L, JT-R, STI-L, and STI-R and HAA, HDSI and HDST are also no greater than 0.8. The correlation coefficients between CEP, CT and AS and the shield attitude are less than 0.8. Pearson coefficients for other input - output pairs are small. Evidently, linear relationships are not clear. This also indicates that traditional mathematical regression models may not be inadequate for describing the complex mapping relationship between shield tunnel parameters and attitude parameters and that using ML models is more conducive to providing accurate shield tunnel attitude prediction.

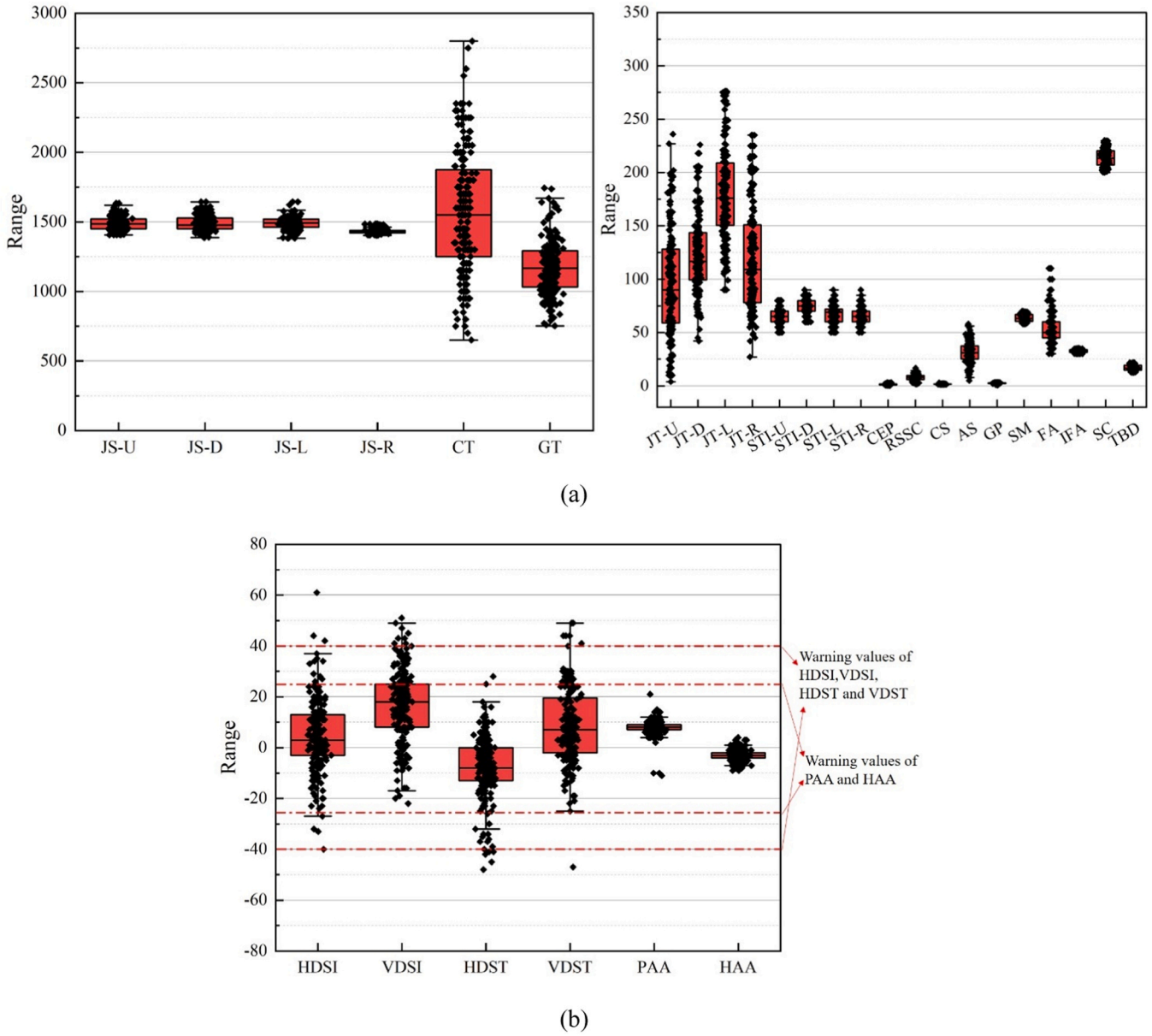


Fig. 7. Inputs and outputs for the shield construction data: (a) Input; (b) output.

4.4. Shield attitude prediction via the PSO-RF

4.4.1. Hyperparameter setting

Optimizing the hyperparameters of the ML model is crucial for improving the accuracy of predictions before applying ML algorithms to predict shield attitudes [55]. Improperly formulated hyperparameter values may increase the number of calculations, prolong the training time and overfit the model. Therefore, it is necessary to set the hyperparameters of the ML model reasonably. When the RF algorithm is employed to train samples, four hyperparameters in the algorithm should be optimized: (1) the number of decision trees (n_estimators), which determines the prediction accuracy and computational complexity of the RF model; (2) the maximum depth of the decision tree (max_depth), which is related to the complexity of the RF model structure and is a key hyperparameter to prevent overfitting of the RF model; (3) the minimum number of samples required for splitting internal nodes (min_samples_split), which prevents the model from being too sensitive to noisy data; and (4) the minimum number of leaf nodes (min_samples_leaf), which controls the purity of leaf nodes and can improve the generalization of the RF model.

To maximize the RF prediction model's accuracy, the PSO fitness is set as the training model's MSE loss function. The particle population number is 20, the maximum number of iterations is 100, the learning factors are both 0.5, and the inertia weight is 1. When the fitness of the particles becomes stable or the number of iterations reaches the maximum, the iterative cycle is terminated, and the optimal hyperparameters are obtained.

Before the parameters are optimized, it is crucial to establish the initial optimization range for the hyperparameters. The setting range of the model hyperparameters depends on the specific research objective. On the basis of the parameter ranges provided in previous studies [29,56,57], combined with specific case backgrounds, the parameter search ranges for the relevant hyperparameters in this study are determined. The initial ranges of n_estimators, max_depth, min_samples_split and min_samples_leaf are [50,150], [20,500], [2,5] and [1,5], respectively. The process of optimizing the RF hyperparameters is depicted in Fig. 9, and the results of hyperparameter optimization are provided in Table 3.

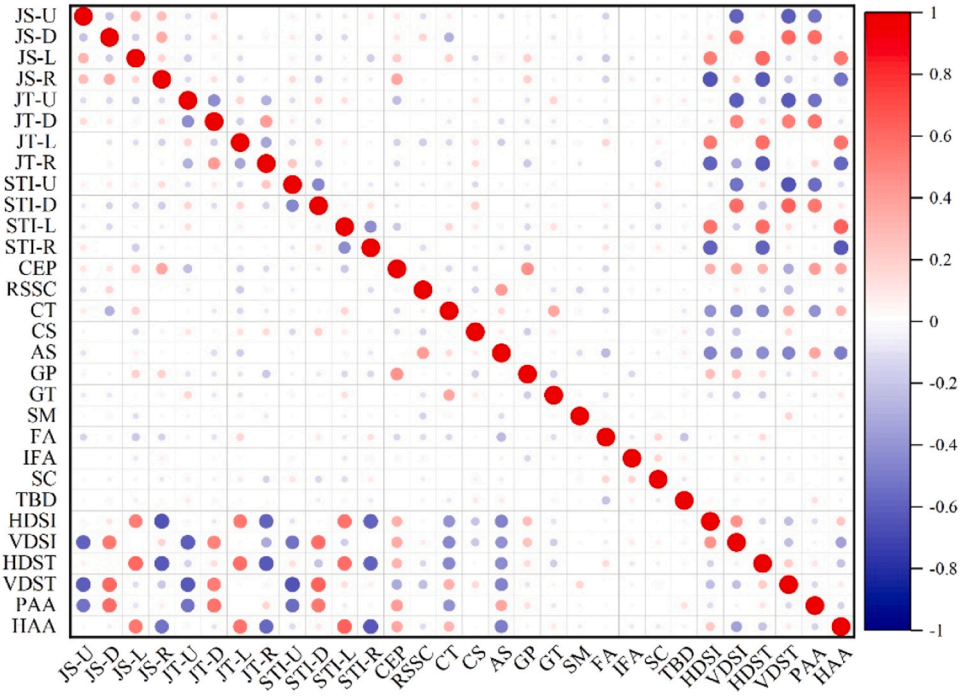


Fig. 8. Heatmap of the Pearson correlation coefficients for the input and output parameters.

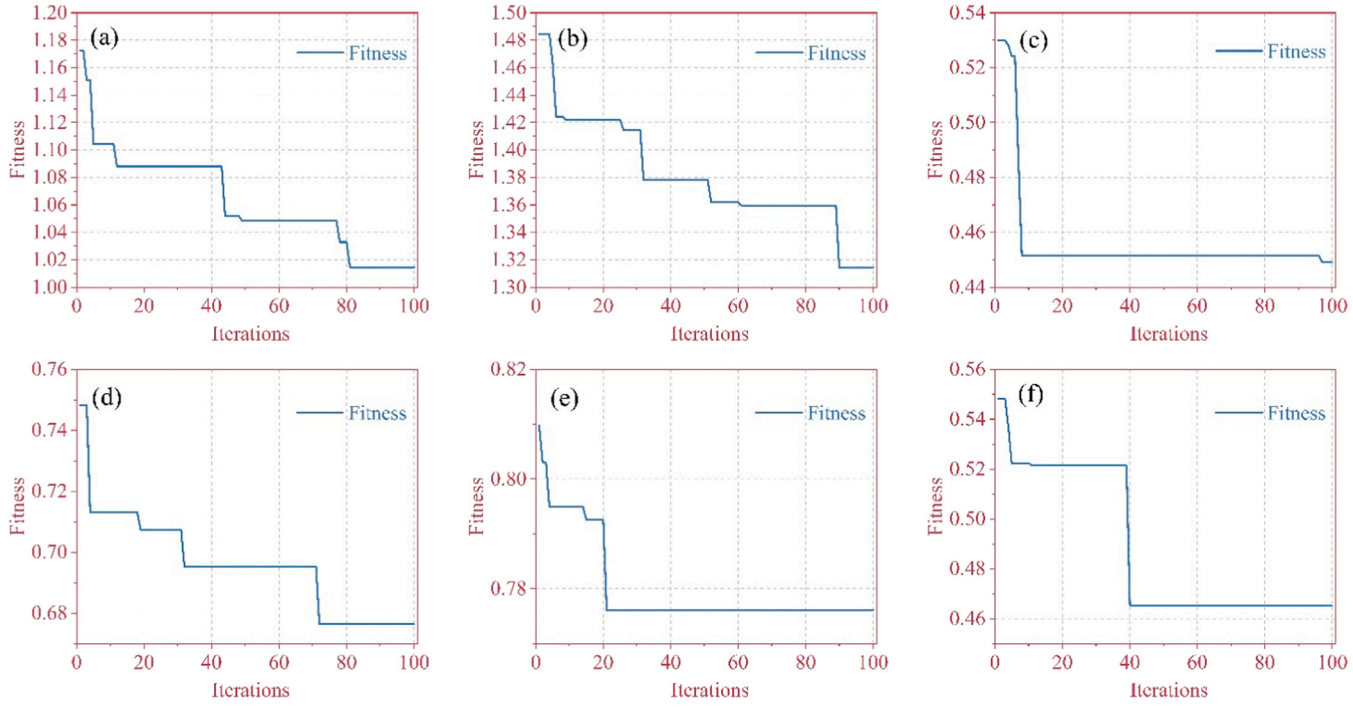


Fig. 9. RF algorithm hyperparameter optimization process: (a) HDSI, (b) VDSI, (c) HDST, (d) VDST, (e) PAA, and (f) HAA.

4.4.2. Model training and performance evaluation

Overfitting is a major challenge in machine learning, often resulting in poor model generalization. To address this, techniques like K-fold cross-validation are employed. In this paper, after hyperparameter optimization, the RF model is trained using 5-fold cross-validation to ensure robust performance across diverse data distributions. Furthermore, the PSO-RF model's performance is closely monitored on both training and test sets, with training halted if the performance gap grows too large, preventing overfitting.

The MSE serves as the evaluation metric, computed for the validation set predictions in each iteration. After repeating this process five times, the average MSE is used as the final performance measure. The model's training error is defined by Eq. (15):

$$CV_{(5)} = \frac{1}{5}\sum_{i=1}^{5} MSE_i \tag{15}$$

where $CV_{(5)}$ is the final training error and MSE_i is the mean square error.

Table 3
Hyperparameter optimization results.

Hyperparameter	Range	Target of shield attitude prediction					
		HDSI	VDSI	HDST	VDST	PAA	HAA
n_eatimators	[50,150]	50	80	50	140	150	100
max_depth	[20,500]	160	35	365	245	325	145
min_samples_split	[2,5]	2	2	2	3	3	4
min_samples_leaf	[1,5]	4	2	3	2	5	4

To evaluate the accuracy of the model, the results of 5-fold CV for six prediction targets using the MAE, RMSE and R^2 are presented in Table 4 and Table 5. Additionally, Fig. 10 shows a comparison between the predicted results and the actual values of the six shield attitude targets.

As shown in Table 4, Table 5 and Fig. 10, the established RF model provides precise predictions of the shield attitude. The prediction errors of the model are low for all the folds. The MAEs of the six shield attitude targets are 2.95 mm, 2.748 mm, 2.814 mm, 3.388 mm, 0.779 mm·m^{-1} and 0.502 mm·m-1. The RMSEs are 3.707 mm, 3.428 mm, 3.576 mm, 4.308 mm, 0.943 mm·m-1, and 0.623 mm·m^{-1}, respectively. The R^2 values are 0.934, 0.943, 0.918, 0.917, 0.916, and 0.928, respectively. The scatterplot in Fig. 10 depicts a well-distributed pattern of points around the fitting line; there are some extreme points, and some deviations are generated because of insufficient data for model learning. These results validate the model's accuracy and robustness, effectively capturing input-output relationships without overfitting or underfitting.

4.5. Optimization and improvement of the shield attitude

4.5.1. SHAP analysis of the predictive models

The influence of input parameters like jack stroke on shield attitude is complex and interrelated, making single-parameter qualitative analysis inadequate for capturing real-world construction dynamics or identifying key factors. To address this, the SHAP algorithm can be employed. By considering the shield attitude as the prediction index, the SHAP algorithm facilitates the evaluation of primary and secondary factors of the shield parameters. This method enables the selection of pertinent shield construction parameters for attitude optimization, providing a robust theoretical foundation for rail transit.

While importance rankings provide insights into the significance of input parameters, they do not inherently reveal the specific direction of influence of each parameter on the model's prediction results. Analyzing importance rankings alone may not fully clarify the decision-making mode of the model. This limitation becomes apparent for shield drivers seeking a deeper understanding of the model's operational rules, creating a gap in trust between humans and the machine model.

SHAP provides another visual interpretation scheme. Features are ordered vertically by importance, with each sample plotted horizontally according to its Shapley value. Values left of the centerline (negative) reduce model output, while those right of the centerline (positive) enhance it. Color intensity reflects feature magnitude, with red indicating higher values and blue lower ones, revealing how each parameter influences predictions. Fig. 11 shows the global summary of the SHAP values and the feature importances of the average SHAP values of the 15 most critical parameters.

According to the global summary diagram of the SHAP values drawn via calculation, JS, JT, STC, CEP, CT, CS and GT exert the most significant influences on the shield attitude prediction results. Therefore, to control the shield attitude, special attention should be given to these parameters.

The results show that (1) the impact of JS on the shield attitude becomes more pronounced with increasing JS. JS-L and JS-R affect mainly the HDSI, HDST and HAA, and JS-L and JS-R have opposite influences. JS-U and JS-D affect mainly the VDSI, VDST and PAA, and JS-U and JS-D have opposite influences. In addition, the impacts of JT and STC on the shield attitude are similar to that of JS. An excessive difference in JS and JT will lead to uneven segment damage, which will seriously interfere with the normal shield attitude during construction. An STC value that is too low during construction can lead to interference between the shield tail and the segment, hindering shield advancement and potentially causing segment dislocation damage. Conversely, excessive STC increases the grouting between the tunnel surface and soil, significantly affecting the load on the surrounding soil. Both scenarios significantly impact the shield attitude. These findings align with prior studies. Zhou et al. proved that JT is positively correlated with the shield tail deviation [58]. Wang et al. identified JS and CEP as key influencing factors in shield attitude deviation [59]. Wada et al. [60] reported that STC significantly impacts the shield attitude.

(2) CEP, CT, CS and GT are the four most important parameters. The same parameter has different effects on different shield attitude targets. The side on which the SHAP value of the GT is less than 0 is represented mainly by red and purple points, indicating that the influence of the GT on the HDSI is reversed; that is, with increasing GT, the smaller the corresponding SHAP value is, the greater the influence on the shield attitude. The influence of CEP on the HDSI is complex; the red and blue points are on the side with SHAP values less than 0, and the purple points are on the side with SHAP values less than 0. The HDSI first increases with increasing CEP; when the HDSI increases within a certain range, it begins to decrease with further increases in CEP. The sample point concentration areas of the other shield parameters are closer to 0 than those of CEP, CT, CS and GT are, indicating that these parameters have a certain impact on the shield attitude, but the impact is small. During construction, a CEP that is too small will cause the tunnel face to collapse and subsequently cause surface settlement, whereas a CEP that is too large will cause ground uplift. Changes in CT and CS cause improper shield movement, resulting in attitude deviation. Adjusting the GT can affect the AS and prevent the attitude instability caused by propulsion being too fast or too slow.

Table 4
Five-fold cross-validation results for HDSI, VDSI and HDST.

Folds	HDSI			VDSI			HDST		
	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2
Fold 1	2.922	3.741	0.934	2.705	3.418	0.943	2.827	3.594	0.917
Fold 2	2.717	3.456	0.943	2.575	3.247	0.949	2.739	3.466	0.923
Fold 3	2.611	3.292	0.949	2.555	3.216	0.950	2.721	3.431	0.925
Fold 4	3.198	3.976	0.925	2.851	3.546	0.939	2.876	3.674	0.914
Fold 5	3.297	4.069	0.922	3.051	3.715	0.933	2.906	3.576	0.912
Average	2.95	3.707	0.934	2.748	3.428	0.943	2.814	3.576	0.918

Table 5
Five-fold cross-validation results for VDST, PAA and HAA.

Folds	VDST			PAA			HAA		
	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2
Fold 1	3.429	4.372	0.915	0.833	0.986	0.909	0.481	0.595	0.935
Fold 2	3.291	4.217	0.920	0.756	0.922	0.920	0.529	0.652	0.922
Fold 3	3.145	4.037	0.927	0.781	0.953	0.915	0.500	0.621	0.929
Fold 4	3.555	4.473	0.911	0.766	0.930	0.919	0.513	0.639	0.925
Fold 5	3.521	4.442	0.912	0.760	0.926	0.920	0.486	0.607	0.932
Average	3.388	4.308	0.917	0.779	0.943	0.916	0.502	0.623	0.928

(3) The five parameters SM, FA, IFA, SC and TBD do not appear among the first 15 most critical parameters; this is caused by the relatively concentrated distributions of SM, FA, IFA, SC and TBD in the sample data. Although the geological conditions and undetermined factors in actual engineering practice have a significant effect on the shield posture, the geological conditions have little effect on the tunnel in this study. This may be caused by the small difference in the relative properties of the soil around the shield in the selected section.

Notably, the SHAP analysis results in this study are similar to the correlation analysis results previously discussed in Section 3.2. This may be related to the stability of the RF model. Since RF consists of multiple decision trees, the interpretability of the model is generally stable, meaning that both the SHAP analysis and Pearson correlation analysis may yield similar rankings of feature importance. Although the results are similar, the goals of SHAP analysis and Pearson correlation analysis differ. Pearson correlation analysis aims to filter shield parameters effectively, whereas the goal of SHAP analysis is to understand the operational mechanism of the PSO-RF model.

4.5.2. Shield attitude optimization control

Based on SHAP and influence rule analyses of input parameters, key shield construction parameters are adjusted for attitude control. The SHAP analysis results indicate that STC significantly impacts shield tunneling attitude. However, STC is a passive parameter rather than an active parameter and cannot be directly controlled. Therefore, the STC is removed as a coupling factor. Parameters such as JS, JT, CEP, CT, CS and GT are adjusted to achieve shield attitude control. The adjustment ranges for these parameters, determined by shield type specifications, are provided in Table 6.

In accordance with GB50446-2017 and engineering practice, the allowable deviation limits for HDSI, VDSI, HDST, and VDST in shield construction are set to $\pm$ 50 mm, whereas for the PAA and HAA, the allowable deviation limit is $\pm$ 30 mm·m^{-1}. As warning thresholds for a construction site, values of 80 % of the specified deviation limits are adopted, resulting in the following limits: HDSI, VDSI, HDST, and VDST should be maintained within $\pm$ 40 mm, whereas PAA and HAA should remain within $\pm$ 25 mm·m^{-1}, prompting corrective adjustments if exceeded. According to the data in Table 2, both PAA and HAA fall within the specified control ranges; therefore, the focus is primarily on controlling HDSI, VDSI, HDST, and VDST. The relevant shield construction parameters are adjusted continually within the stipulated adjustment ranges. The RF model is then employed to predict the shield attitude control targets until they align with the specified control criteria. The outcomes are presented in Fig. 12.

Fig. 12 shows the effective control of the shield attitude following the adjustment of the relevant construction parameters. The four shield attitude targets are maintained within the warning range without any exceedances. This method is applied to Guiyang Rail Transit Line 3, and its effectiveness is shown by comparing the shield attitudes before and after the control adjustments, as depicted in Fig. 13.

The primary factor influencing shield attitude is JS. To elaborate on the shield attitude adjustment process, this method's fundamental principle is demonstrated by adjusting the JS to change the VDSI. The JSs on the shield are divided into four groups: upper, lower, left and right. For the four jacks in each group, Fig. 14(a) illustrates the predicted VDSI values in the tunneling process for rings 201–400, and Fig. 14(b) shows the difference between JS-U and JS-D. The downward deviation of the shield can be corrected by setting JS-U to be greater than JS-D; similarly, JS-U can be set to less than JS-D to correct the upward deviation of the shield.

To illustrate this rectification process with the example of the VDSI at ring 305, the following specific steps are used: upper and lower limits of −150 mm to 150 mm are set for the difference between JS-U and JS-D. The initial difference between JS-U and JS-D is set as 0, with 37.5 mm as the gradient. Considering the VDSI under nine conditions (namely, −150 mm, −112.5 mm, −75 mm, −37.5 mm, 0 mm, 37.5 mm, 75 mm, 112.5 mm, and 150 mm), the difference between JS-U and JS-D corresponding to the minimum VDSI is found. In the interval [-150 mm, 150 mm], the shield offset can be reduced by modifying the difference between JS-U and JS-D. When the difference between JS-U and JS-D is 150 mm, the maximum VDSI is 53 mm; when the difference between JS-U and JS-D is −150 mm, the minimum VDSI is 12 mm. Therefore, by adjusting the difference between JS-U and JS-D, the VDSI can be effectively reduced. Maintaining a good shield attitude can reduce soil overdigging, excessive friction and extrusion of the shield shell into the soil, thus reducing soil disturbance during tunneling and minimizing the amount of surface settlement. When the shield attitude is offset, it needs to be adjusted. In accordance with this same principle, adjustments to the difference between the upper and lower jacks can be made to rectify the deviation. It is essential to exercise caution and avoid excessively large corrections, as these may lead to issues such as segment rupture. In summary, when shield attitude deviation occurs during the TBM tunneling process, adjusting construction control parameters significantly affecting the shield attitude is an effective means of correction.

4.6. Twin shield attitude control system

A twin shield attitude control system, enhanced by artificial intelligence, offers a visual decision-making environment for shield attitude control. Its framework comprises four core layers: the physical entity layer, data layer, digital twin layer, and service layer, as depicted in Fig. 15.

The physical entity layer comprises the shield construction equipment, driving process stages, and data acquisition system. The TBM is categorized on the basis of the distinct functions of its equipment, which include the cutter drive system, shield system, propulsion and articulation system, muck conveying system, and other auxiliary systems. The parameters and performance of the relevant systems vary in different tunneling stages. Real-time data collection is essential for interfacing with the digital twin virtual model, as this ensures consistency throughout the tunneling process. The data acquisition system includes a shield navigation system and a shield parameter collection system. It can not only sense the operating status of the driving system and collect real-time data on the driving system load, advance speed and chamber earth pressure but also actively engage with system data for dynamic model updates and real-time adjustment of the driving system parameters.

The data layer wirelessly transmits the gathered twin data to the data

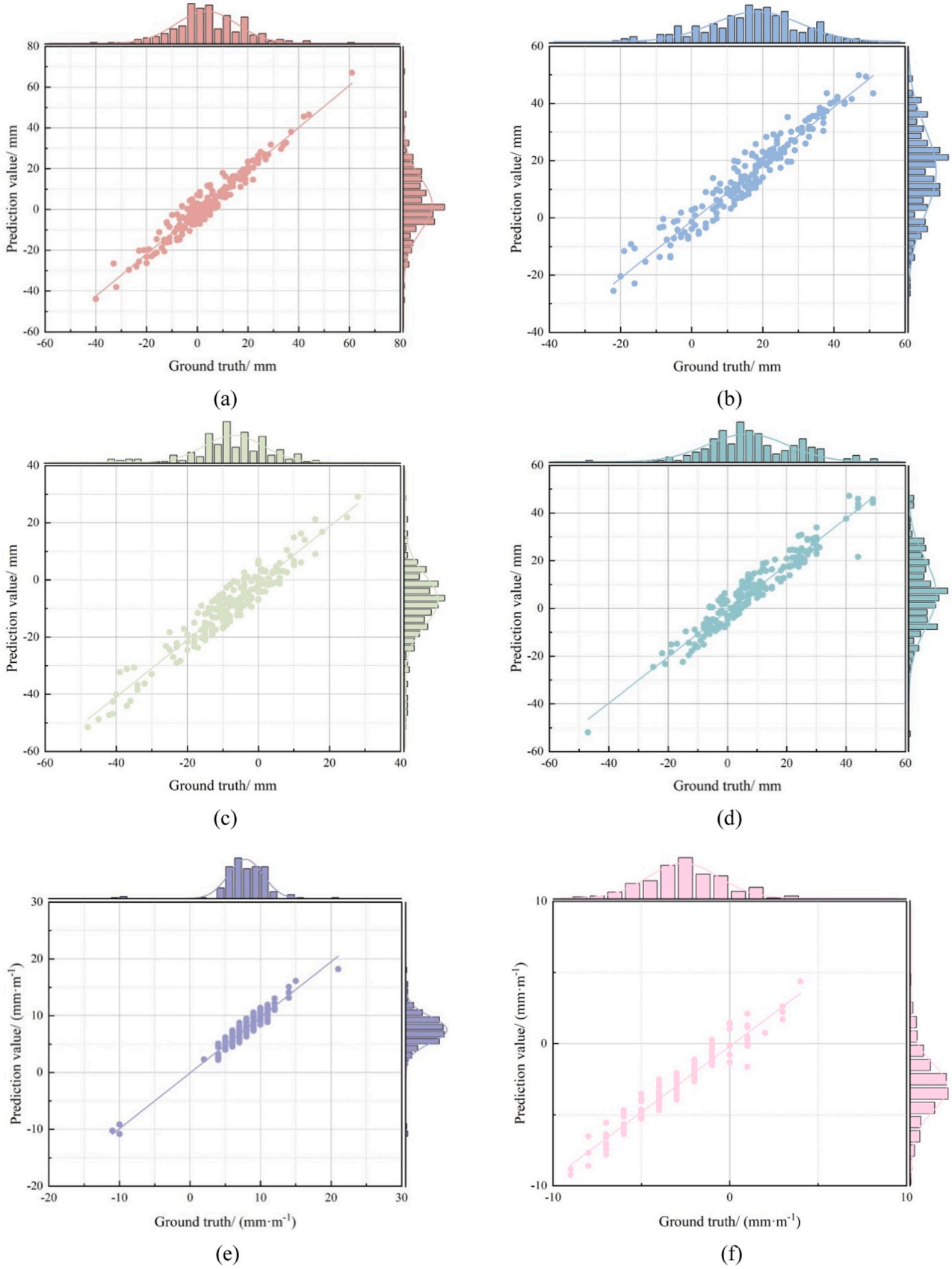


Fig. 10. Comparison of the predicted and real values of six shield attitude targets: (a) HDSI; (b) VSI; (c) HDST; (d) VTS; (e) PAA; (f) HAA.

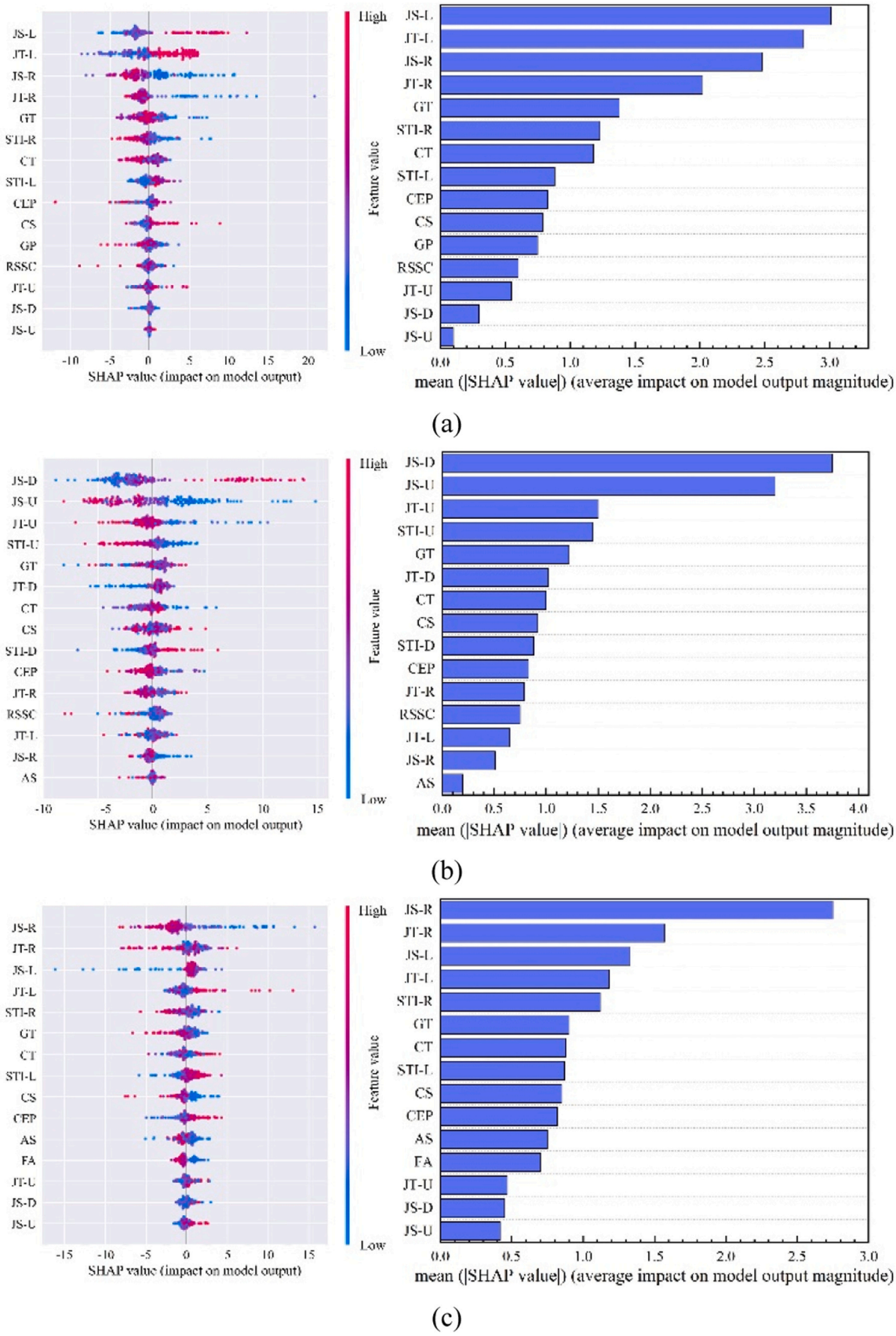
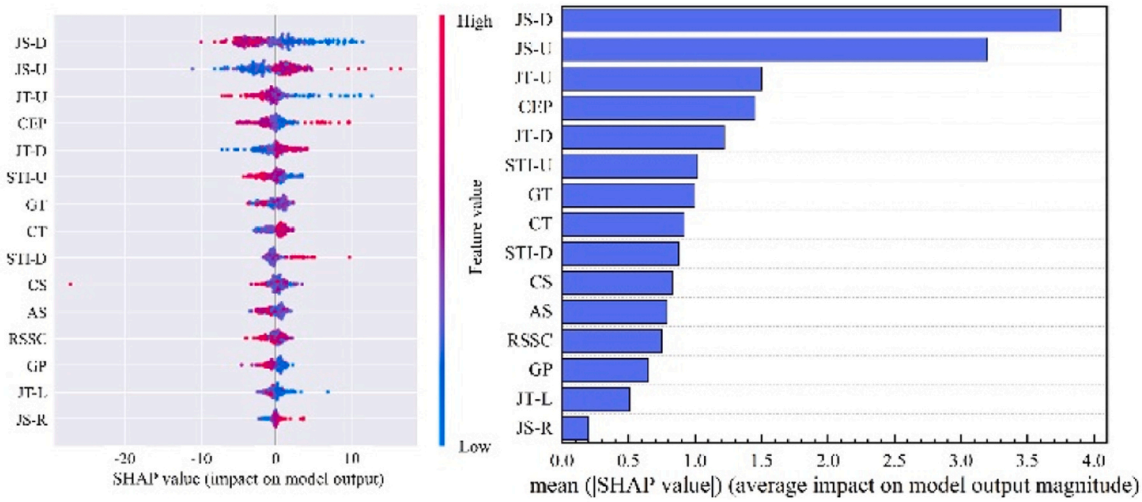
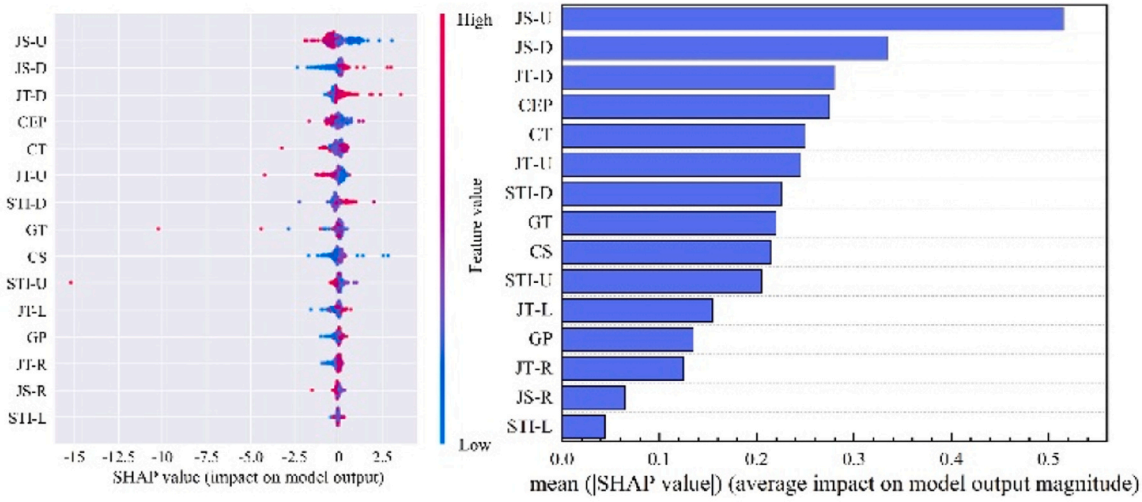


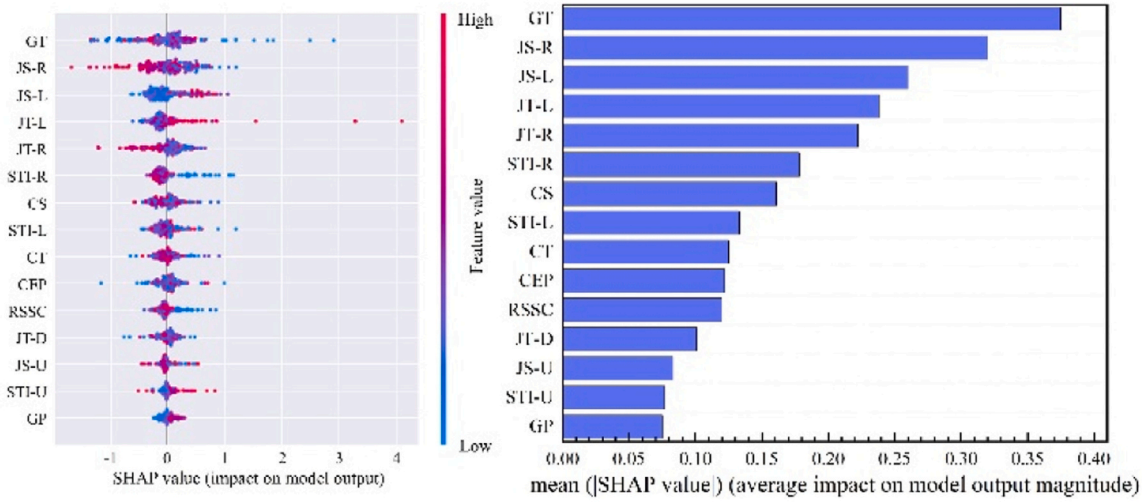
Fig. 11. Global summary of SHAP values and feature importances of average SHAP values: (a) HDSI; (b) VDSI; (c) HDST; (d) VDST; (e) PAA; (f) HAA.



(d)



(e)



(f)

Fig. 11. (*continued*).

Table 6
Parameter adjustment ranges.

Adjustment parameters	Parameter ranges	Adjustment parameters	Parameter ranges
JS-U/mm	0–2000	JT–L/mm	0–300
JS-D/mm	0–2000	JT-R/mm	0–300
JS–L/mm	0–2000	CEP/bar	0–3
JS-R/mm	0–2000	CT/(kN·m)	0–3000
JT-U/mm	0–300	CS/(r·min^{-1})	0–3
JT-D/mm	0–300	GT/kN	0–10000

processing server for further processing, analysis, and feedback through the database. The twinning data of the shield attitude control system constitute an integrated module of all the data under the twinning framework; these data include the device data of the shield attitude control system, virtual model data, shield attitude prediction and control data and interactive fusion data. The equipment data of the shield attitude control system mainly include the geometric and physical performance data of the shield, as well as real-time mechanical data and equipment operation data during the driving process. The virtual model data of the shield attitude control system include the simulation and updating data of the twin model, which represent the shield attitude control system's operation within the virtual space. The shield attitude prediction and control data are mainly composed of shield attitude prediction data, construction parameter adjustment data and decision data, which can interact with physical entities and virtual models in real time. The interactive fusion data are the data generated after mining and processing the above three datasets, such as via outlier and missing value processing, data dimensionality reduction, and feature extraction; this facilitates real-time interaction among the physical entity, virtual model, and service applications of the shield.

In the digital twin layer, a virtual model of the shield is established using the collected data. This virtual model not only captures the geometric dimensions and material characteristics of the physical shield but also continually monitors the real-time tunneling activities of the machine. Real-time dynamic simulations and interactions encompassing

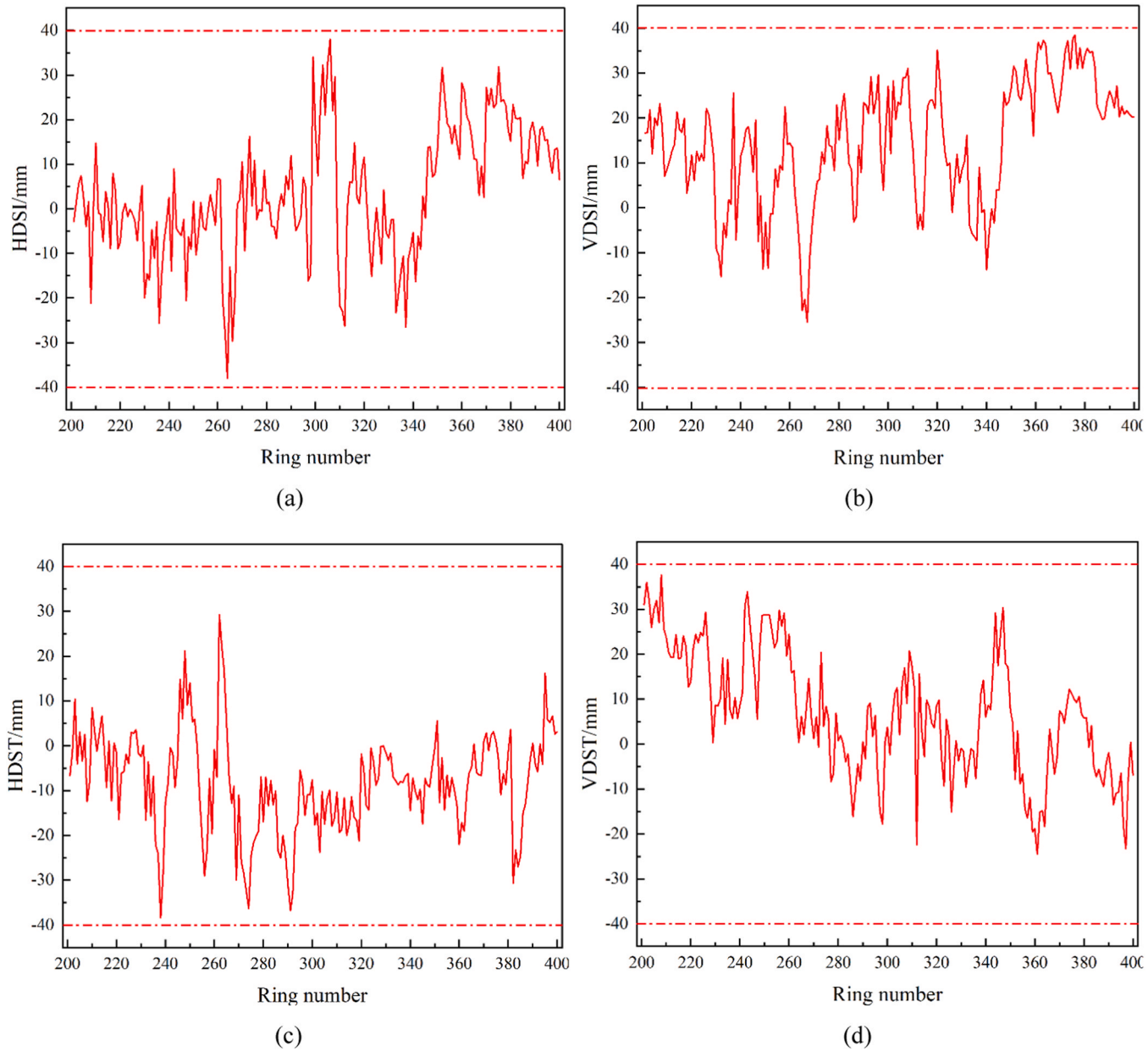


Fig. 12. Shield attitude control results after adjusting the parameters for shield construction control: (a) HDSI; (b) VDSI; (c) HDST; (d) VDST.

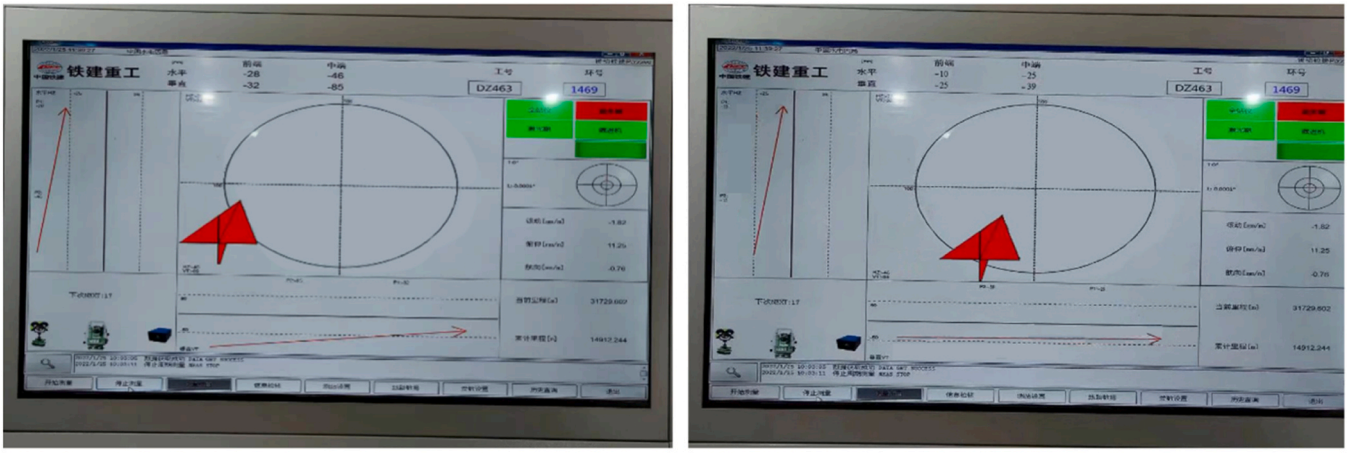


Fig. 13. Shield attitudes before and after control adjustment: (a) Before adjustment; (b) after adjustment.

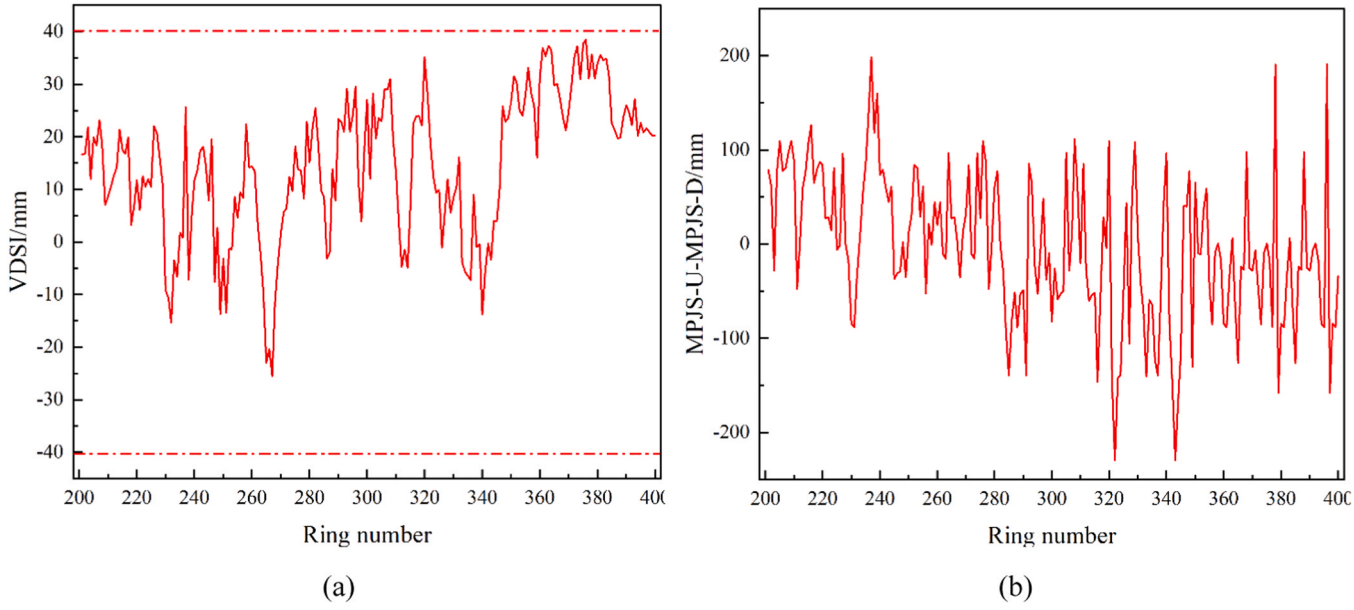


Fig. 14. Comparison between the shield attitude and construction control parameters: (a) The predicted HDST during tunneling; (b) the difference between MPJS-L and MPJS-R.

the four dimensions of geometry, physics, behavior, and rules are conducted. This virtual model maintains a bidirectional interaction channel with the physical entity. his connection allows real-time data synchronization, continuous model updates, and feedback from ML-based predictions to the physical entity.

In the service layer, the virtual model can simulate the shield attitude in a real-time, comprehensive virtual space. This approach empowers managers to proactively identify attitude deviations during the tunnel shield tunneling process, enabling them to issue construction instructions and guidance to onsite personnel. This approach also facilitates the optimization and adjustment of shield construction parameters, leading to well-informed decision-making measures.

This paper proposes a twin shield attitude control system, integrating shield attitude adjustment control processes with digital twin technology to enhance the existing shield operating system through attitude warning and parameter inference functions. As shown in Fig. 16, the system interface comprises four main sections: the upper left displays actual and recommended JS values in four directions; the lower left shows shield attitude monitoring values, with four disks indicating real-time shield attitude and four frames tracking 24-h trends; the upper right presents the current construction section plan, while the lower right provides a real-time visual display of shield attitude. A color-coded system is employed to indicate the directions of key shield attitude parameters: a blue arrow is used for HDSI and HDST when their values are positive, a yellow arrow is used for VDSI and VDST when their values are positive, and a green arrow signifies the direction of PAA and HAA when their values are positive.

5. Discussion

The prediction accuracy and computational efficiency of ML models are critical to the DT platform, forming the foundation for effective shield attitude control. To validate the integration of PSO-RF models into the DT system, a comparative analysis was conducted, evaluating different ML model combinations and hyperparameter optimization methods.

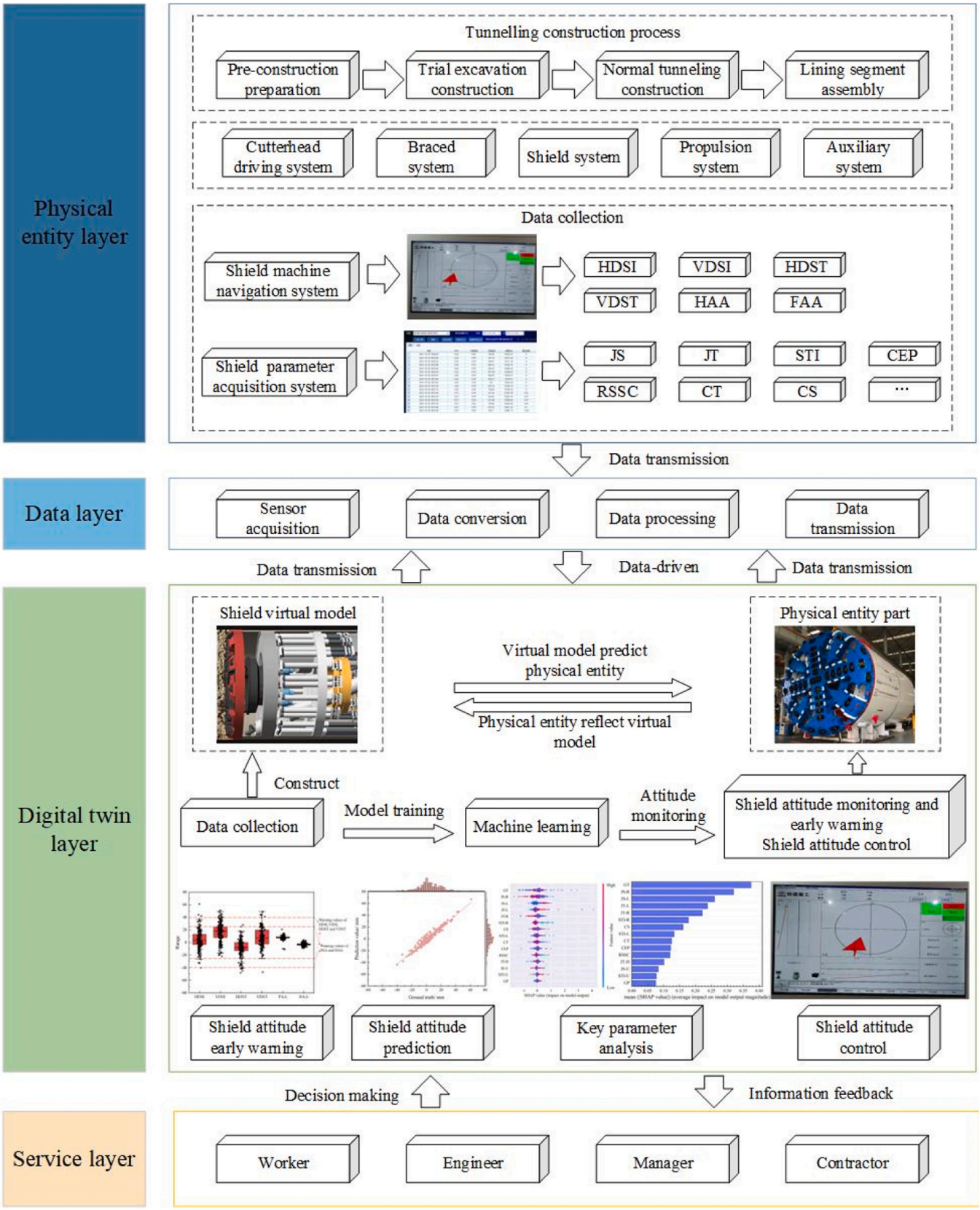


Fig. 15. Framework of the twin shield attitude control system.

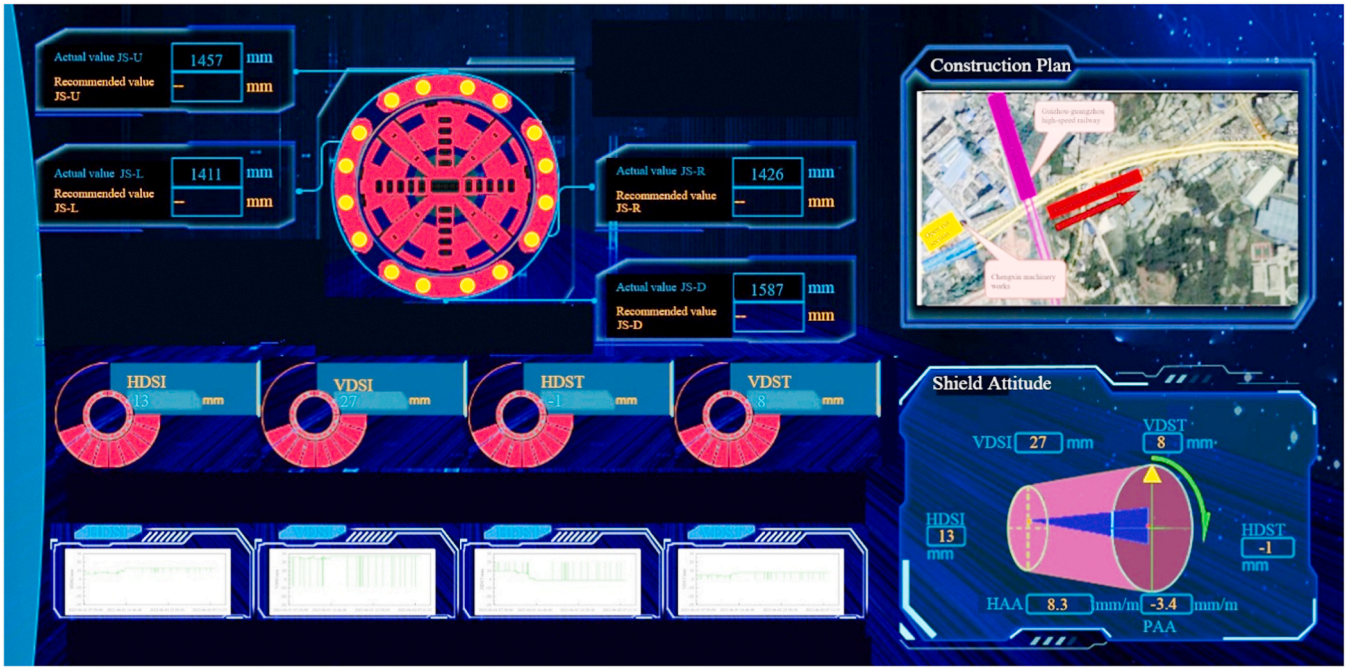


Fig. 16. Twin shield attitude control system.

5.1. Comparison of prediction methods

To validate the effectiveness of the proposed PSO-RF model, it is compared with several advanced prediction models, including XGBoost, LightGBM, GRU, and LSTM, which have demonstrated strong predictive capabilities in prior research. While LSTM and GRU represent advanced deep learning techniques, XGBoost and LightGBM are high-performance tree-based ensemble methods, each demonstrating strong predictive capabilities in prior research [22–24,27]; the constructed PSO-RF model is compared with these models in this section.

Hyperparameter optimization significantly impacts ML model performance by enhancing prediction accuracy. The hyperparameters optimized by the XGBoost and LightGBM algorithm models are shown in Table 6. GRU and LSTM have four important hyperparameters: hidden_size, learning_rate, num_layers and activation functions.

These algorithms have the same dataset, with the PSO algorithm optimizing their hyperparameters based on the study's sample size and feature dimensions, following established PSO parameter settings from prior literature. The model training adheres to the methodology outlined in Section 3.4.1. Beyond prediction accuracy, computational efficiency is evaluated, with Table 7 detailing each model's training and prediction times, while Table 8 presents the optimized hyperparameters. These hyperparameters are then used to build predictive models for the six shield attitude targets, with model performance assessed via five-fold cross-validation using R^2, RMSE, and MAE metrics. Table 9 presents the predictions of the five latest methods used for comparison. The predictive performance of an ML model needs to be evaluated via multiple metrics. To visually compare the predictive performance of different ML models in various dimensions, Taylor diagrams are used. Taylor diagrams can display three-dimensional information on a two-dimensional plane, which is challenging to achieve with traditional scatterplots or bar charts. This allows an intuitive and easily understandable comparison of the performance between different models, which helps identify the model with the best performance. In the Taylor diagram, the correlation coefficients, standard deviations, and RMSEs between the prediction results of the five different models and the monitoring data are presented. Fig. 17 shows the Taylor plots of the predictions of the five models. These results are comprehensively evaluated and analyzed below.

(1) RF produces more accurate predictions than do the existing algorithms. In Table 9, RF has the highest accuracy in predicting HDSI, VDSI, HDST, PAA and HAA, with the smallest MAE and RMSE and the largest R^2. Taking HDSI as an example, the RF reduces MAE by 12.58 % ((3.321–2.95)/2.95), 3.29 %, 6.92 %, and 9.42 % compared to XGBoost, LightGBM, GRU, and LSTM, respectively, while lowering RMSE by 12.09 % ((4.155–3.707)/3.707), 6.10 %, 9.17 %, and 10.47 % and improving R^2 by 1.71 % ((0.934–0.918)/0.918), 0.86 %, 1.28 %, and 1.50 %. The Taylor diagram in Fig. 17(a)~(f) illustrates each model's predictive accuracy based on its proximity to the reference point, with RF being the closest, followed by LightGBM, GRU, LSTM, and XGBoost, demonstrating that the RF model delivers superior performance and higher precision compared to the latest methods.

(2) The ensemble learning methods (RF, XGBoost and LightGBM) have better predictive power than the deep learning methods (LSTM and GRU). The average prediction accuracies of the deep learning and ensemble learning methods are shown in Table 9. Taking HDSI as an example, the MAE of ensemble learning is 2.74 % lower than deep learning's ((3.191–3.106)/3.106), RMSE is 3.54 % lower ((4.071–3.933)/3.933), and R^2 is 0.54 % higher ((0.926–0.921)/0.921). Although deep learning is strong in mapping nonlinear variable relationships, ensemble learning is more effective for predictive modeling with relatively small data samples. Thus, ensemble learning is more suitable for constructing the predictive model in this situation.

Table 7
Runtimes of the five algorithms.

Model	PSO-RF	PSO-XGBoost	PSO-LightGBM	PSO-GRU	PSO-LSTM
Runtime (s)	321.18	372.84	359.26	891.85	912.89

5.2. Comparison of hyperparameter optimization methods

The hyperparameter optimization method and operational efficiency of ML models are important aspects worth discussing, as they play crucial roles in the service application layers of subsequent DT systems. Random search (RS), grid search (GS), and Bayesian optimization (BO)

Table 8
Hyperparameter optimization results for the five algorithms.

Algorithm	Hyperparameter	Optimization range	Target of shield attitude prediction					
			HDSI	VDSI	HDST	VDST	PAA	HAA
XGBoost	colsample_bytree	[0.5, 0.8]	0.70	0.76	0.54	0.76	0.57	0.52
	gamma	[0,5]	2	3	2	2	1	3
	learning_rate	[ln0.3, ln0.6]	0.600	0.598	0.600	0.599	0.599	0.590
	max_depth	[4,20]	8	14	4	20	12	4
	max_leaves	[0,150]	50	50	150	100	150	100
	min_child_weight	[1,5]	2	4	3	1	5	2
	reg_alpha	[0,1]	0.97	0.32	0.85	0.76	0.29	0.48
	reg_lambda	[0,1]	0.46	0.17	0.60	0.02	0.52	0.26
LightGBM	colsample_bytree	[0.6, 1]	0.78	0.77	0.88	0.81	0.86	0.98
	learning_rate	[lg0.005, lg0.5]	0.05	0.11	0.04	0.21	0.07	0.03
	min_child_samples	[20,500]	20	60	25	20	35	390
	num_leaves	[20,150]	43	74	88	91	32	117
	reg_alpha	[0,1]	0.33	0.66	1.00	0.92	0.76	0.84
	reg_lambda	[0,1]	0.91	0.61	0.10	0.13	0.41	0.25
	subsample	[0.5, 1]	1	0.71	1	0.86	0.92	0.70
	subsample_for_bin	[20,000,300,000]	160,000	120,000	100,000	220,000	20,000	160,000
GRU	hidden_size	[10,20]	16	18	20	18	16	20
	learning_rate	[ln0.001, ln0.012]	0.007	0.003	0.010	0.003	0.001	0.012
	num_layers	[1,3]	2	2	3	1	1	2
	Activation functions	ReLU, ELU	ELU	ReLU	ELU	ELU	ReLU	ELU
LSTM	hidden_size	[1,10]	2	8	4	8	10	6
	learning_rate	[ln0.001, ln0.012]	0.002	0.003	0.003	0.001	0.007	0.001
	num_layers	[1,3]	3	2	2	3	1	1
	Activation functions	ReLU, ELU	ELU	ELU	ELU	ELU	ReLU	ELU

Table 9
Results predicted by the five compared methods.

Prediction accuracy	Algorithm		Target of shield attitude prediction					
			HDSI	VDSI	HDST	VDST	PAA	HAA
MAE	Ensemble learning	RF	2.950	2.748	2.814	3.388	0.779	0.502
		XGBoost	3.321	3.257	3.514	3.757	0.820	0.536
		LightGBM	3.047	2.872	3.092	3.530	0.794	0.515
		Average	3.106	2.959	3.140	3.558	0.798	0.518
	Deep learning	GRU	3.154	2.940	3.196	3.659	0.796	0.525
		LSTM	3.228	3.094	3.263	3.697	0.813	0.531
		Average	3.191	3.017	3.229	3.678	0.804	0.528
RMSE	Ensemble learning	RF	3.707	3.428	3.576	4.308	0.943	0.623
		XGBoost	4.155	3.847	3.888	4.667	1.024	0.676
		LightGBM	3.933	3.582	3.683	4.405	0.972	0.643
		Average	3.932	3.619	3.716	4.469	0.980	0.647
	Deep learning	GRU	4.047	3.699	3.787	4.546	0.994	0.653
		LSTM	4.095	3.773	3.813	4.583	1.014	0.668
		Average	4.071	3.736	3.800	4.564	1.004	0.661
R^2	Ensemble learning	RF	0.934	0.943	0.918	0.917	0.916	0.928
		XGBoost	0.918	0.928	0.903	0.903	0.901	0.916
		LightGBM	0.926	0.937	0.913	0.913	0.911	0.923
		Average	0.926	0.936	0.911	0.911	0.910	0.922
	Deep learning	GRU	0.922	0.933	0.908	0.908	0.907	0.921
		LSTM	0.920	0.931	0.907	0.906	0.903	0.917
		Average	0.921	0.932	0.907	0.907	0.905	0.919

have been widely applied in hyperparameter optimization of ML models and have achieved good application results [23,61,62]. To prove the rationality of using PSO for RF algorithm optimization in this study, PSO is compared with RS, GS, and BO in optimizing RF model hyperparameters, assessing their impacts on ML model performance and operational efficiency. Before conducting the comparison, it is necessary to set the search ranges of the hyperparameters. For fairness, Table 3 lists the same optimization parameters and search ranges for all these methods. The prediction accuracies and runtimes of different optimization methods, presented in Table 10 and Table 11, are comprehensively evaluated and analyzed below.

Using PSO to optimize the hyperparameters of RF can achieve a balance between runtime and accuracy, making the integration of the PSO-RF model in the DT platform reasonable. Among all the models, RS-RF has the shortest runtime, whereas BO-RF achieves the highest average predictive accuracy. The runtime of PSO-RF is improved by 7.48 % compared with that of RS-RF, with an average R^2 increase of 2.67 %. Compared with that of BO-RF, the runtime of PSO-RF is reduced by 32.78 %, but the average R^2 decreases by only 0.75 %. In the DT platform, to enable timely and intelligent control of the TBM shield posture, the integrated model needs to achieve a balance between accuracy and runtime. The results show the PSO - RF model has an acceptable runtime and good predictive performance.

6. Limitations and future work

While the shield attitude dynamic prediction and control method proposed in this paper has shown significant progress in real-world applications, it still has certain limitations and potential areas for improvement that warrant further investigation.

1. The proposed shield attitude prediction method focuses on key influencing parameters, but its current sample data is limited to

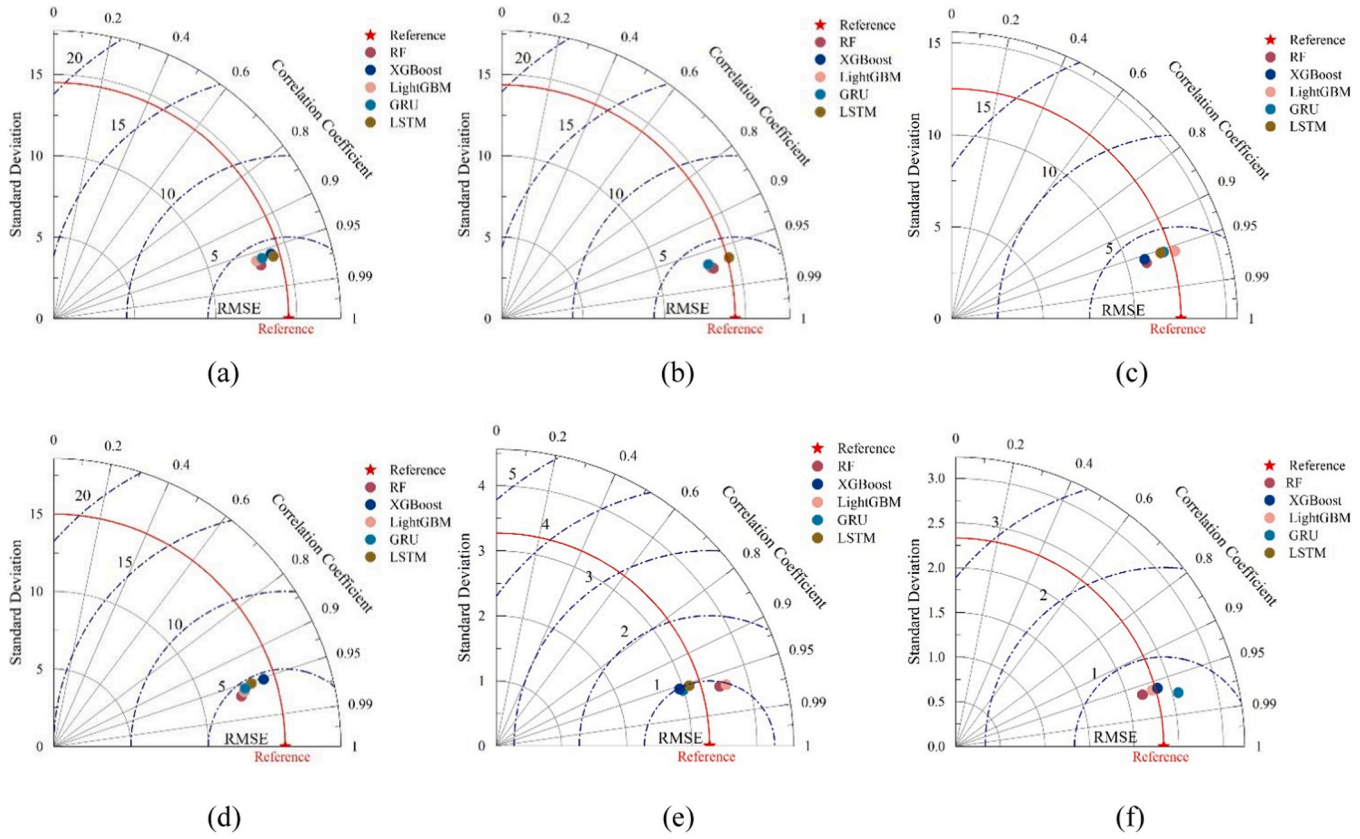


Fig. 17. Taylor plots of the predicted results of the five models: (a) HDSI; (b) VDSI; (c) HDST; (d) VDST; (e) PAA; (f) HAA.

Table 10
Runtimes of different hyperparameter optimization methods.

Methods	RS-RF	GS-RF	BO-RF	PSO-RF
Runtime (s)	298.84	384.12	477.82	321.18

individual regional projects. Future research should use more engineering samples, a larger sample size, and expanded ranges of tunnel burial depths, stratum parameters [63] and other factors to promote broader applications of the proposed method.

2. While the proposed method has proven effective in the Guiyang Metro project, its broader applicability requires further validation across diverse geological conditions, as regional variations in China and worldwide may impact shield attitude prediction and control. Future research should establish a more comprehensive system that accounts for geology-specific influencing factors to enable accurate shield attitude prediction and automatic control.

3. Relevant physical mechanisms should be further considered in the proposed method. Although the proposed PSO-RF algorithm achieved good application results in the current study, the lack of relevant physical mechanisms is an important factor that may hinder its generalizability. Future research should explore suitable means to incorporate physical mechanisms into the prediction model and thus further improve the model's ability to cope with challenges such as geologic uncertainty.

7. Conclusions

This paper aims to supplement and enhance the existing shield operation systems to achieve dynamic prediction and control of the shield attitude during TBM excavation. We propose a method that combines high-performance ML models with a DT platform, which can assist with tunnel construction in practical engineering and strengthen the interaction between the construction site and intelligent algorithms. The proposed method achieves accurate prediction of the shield attitude

Table 11
Prediction performance of different hyperparameter optimization methods.

		HDSI	VDSI	HDST	VDST	PAA	HAA	Average
RS-RF	R^2	0.913	0.917	0.874	0.903	0.899	0.903	0.902
	RMSE	4.271	3.901	4.051	4.657	1.048	0.681	3.102
	MAE	3.415	3.107	3.484	3.748	0.851	0.549	2.526
GS-RF	R^2	0.917	0.922	0.909	0.912	0.911	0.922	0.916
	RMSE	4.252	3.721	3.794	4.401	0.975	0.641	2.964
	MAE	3.299	2.985	3.188	3.525	0.807	0.524	2.388
BO-RF	R^2	0.925	0.928	0.941	0.944	0.933	0.929	0.933
	RMSE	3.941	3.682	3.214	3.887	0.878	0.619	2.704
	MAE	3.088	2.861	2.662	3.014	0.695	0.494	2.136
PSO-RF	R^2	0.934	0.943	0.918	0.917	0.916	0.928	0.926
	RMSE	3.707	3.428	3.576	4.308	0.943	0.623	2.764
	MAE	2.950	2.748	2.814	3.388	0.779	0.502	2.197

through the proposed PSO-RF model and identifies key shield parameters using SHAP analysis. Accordingly, 80 % of the extreme control value for shield attitude deviations specified in the relevant regulations is set as the early warning threshold for the construction site. By continuously adjusting key shield parameters within the specified range, the shield attitude can be maintained within the established control standards.

The case examined in this paper is from Guizhou Province, China, which is the core area of karst landforms in Southwest China. The concealment and uncertainty of karst caves in karst landforms are the main obstacles to tunnel construction. According to the results of this study, the construction party achieved one year ahead of schedule in line 3, which demonstrates the validity and feasibility of the proposed methodology. The results reveal that the following: (1) The proposed PSO-RF method can reliably predict the shield attitude. The MAE, RMSE, and R^2 range from 0.502 to 3.388, 0.623–4.308, and 0.916–0.943, respectively, for the six shield attitude prediction targets. (2) The SHAP interpretability analysis reveals that the seven most influential shield construction parameters regarding the shield attitude are JS, JT, STC, CEP, CT, CS and GT. These parameters therefore need to be strictly controlled. (3) According to the shield attitude adjustment control process and digital twin technology, a twin shield attitude control system is designed to supplement and enhance the existing shield operating system, which includes an attitude warning function and a parameter inference function.

CRediT authorship contribution statement

Yang Liu: Writing – review & editing, Visualization, Resources. **Tiejun Li:** Writing – original draft, Formal analysis, Data curation. **Jun Liu:** Methodology, Investigation. **Xianguo Wu:** Project administration, Funding acquisition, Conceptualization. **Feiming Su:** Validation, Supervision, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1
Pseudocode of PSO-RF

```
    Define: PSO parameters (number of particles N, maximum number of iterations MaxIter, inertia
weight w, learning factors C1, C2)
    Define: The hyperparameter list params that needs to be optimized for the RF model
    Initialize particle swarm particles [N], with each particle having a position and velocity
    for iteration = 1 to MaxIter do
        for particle in particles do
            RFModel = RandomForest(params=particle.position)   // Train an RF model using the current
        particle position (hyperparameters)
            score = evaluateModel(RFModel) // Evaluation model
            if score > particle.pbestScore then   // Update personal best value
                        particle.pbestScore = score
                        particle.pbestPosition = particle.position
            if score > gbestScore then   // Update global best value
                        gbestScore = score
                        gbestPosition = particle.position
        end for
    for particle in particles do
            particle.velocity=w*particle.velocity+c1*rand()*(particle.pbestPosition-
        particle.position) + c2 * rand() * (gbestPosition - particle.position)

            particle.position = particle.position + particle.velocity
    end for
End for
```

Appendix B. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.asoc.2025.113294.

Data availability

Data will be made available on request.

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